GS EXTERIOR EXPERTS · DECK DIVISION · PROPOSAL

Elevated Deck · 5549 S Lansing Ct

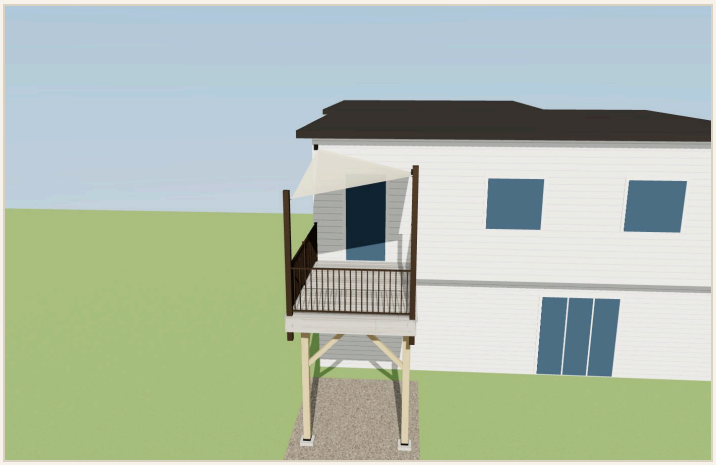
Prepared for Rachel Chiu · 5549 S Lansing Ct, Englewood, CO 80111 · October 19, 2026

CHECK OR ACH	FINANCED	MONTHLY
$22,173	**$23,842**	**$277**
	no money down	6.99% · 10 years



A new, larger deck off your back door: 8' along the house and 12' out, 10' above the backyard, in place of the existing one. Framed in TrueFrame lumber at 16" on center on a doubled 4x10 Douglas fir beam and two 6x6 Douglas fir #1 posts with knee braces, on Diamond Pier foundations, with joist tape on every member. TimberTech Reserve in Driftwood with a single picture frame on the three open sides and CONCEALoc hidden fasteners, so no screw heads show. TimberTech Impression Rail Express in Bronze with vertical balusters and the modern top rail. A tensioned four-corner shade sail over the deck, anchored to the house under the eave and to two steel posts that double as the front corner posts of the railing. The house attachment is engineered.

AT A GLANCE



From the yard



Isometric from the open corner

Deck	92 SF · 10'-0" above grade · 7'-11 1/8" x 11'-6 9/16" over fascia
Decking	TimberTech Reserve Driftwood · CONCEALoc fasteners
Rail	TimberTech Impression Rail Express, vertical balusters, modern top rail, Bronze, 36" · 31 LF
Structure	2x10 pressure-treated frame at 16" on centre, a doubled 4x10 Douglas fir beam; 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50
Design load	50 psf (40 psf live governs), stamped by a Colorado engineer

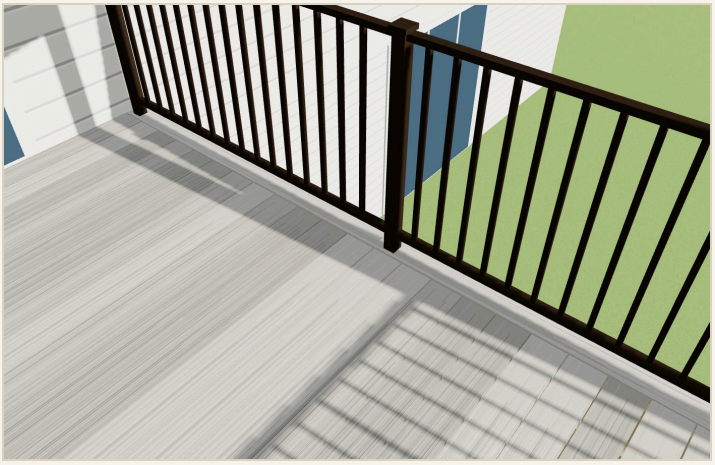
WHAT IT IS included in the price

ELEMENT	SELECTION
Decking	TimberTech Composite Reserve, Driftwood. Grooved boards, 3/16" gaps, matching fascia. Class C, not WUI; 30-yr product / 30-yr fade & stain.
Railing	TimberTech Impression Rail Express, vertical balusters, modern top rail, Bronze, 36". 31 LF.
Frame	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"; 2-ply 2x10 front and sides; a doubled 4x10 Douglas fir beam.
Foundation	2 Diamond Pier foundations (DP-50/50) — no excavation.
Design load	30 psf snow, stamped by a Colorado engineer.
Hardware	Simpson ZMAX galvanized connectors. CONCEALoc fasteners.
Site	The existing deck removed and hauled off.

RENDERS 3D model · design intent



From the front corner



Standing on the deck



Low isometric



Under the deck — posts, beams, footings



Jul 15 · 9 AM — sun E, 35° up



Jul 15 · 12 PM — sun SE, 67° up



Jul 15 · 3 PM — sun WSW, 60° up

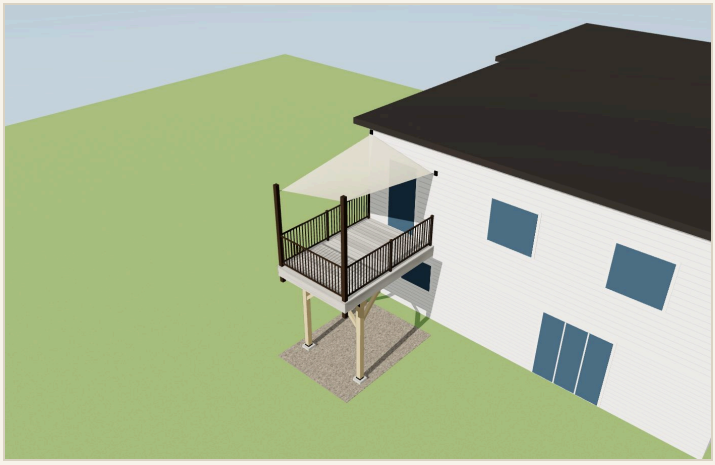


Jul 15 · 6 PM — sun W, 26° up

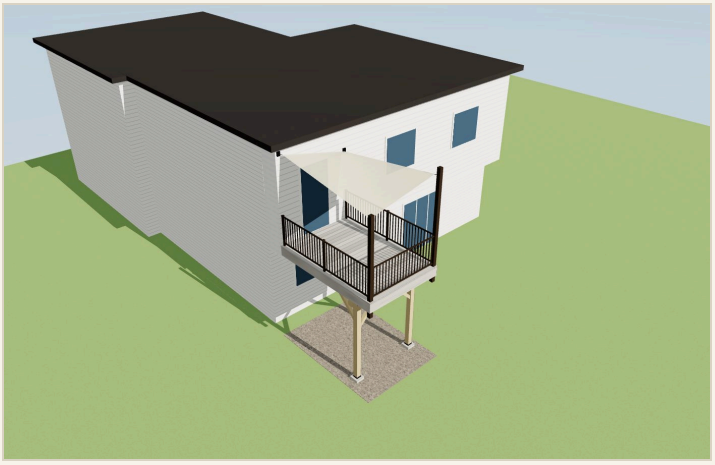


Dec 21 · 3 PM — sun SW, 14° up

ISOMETRICS the model from four corners



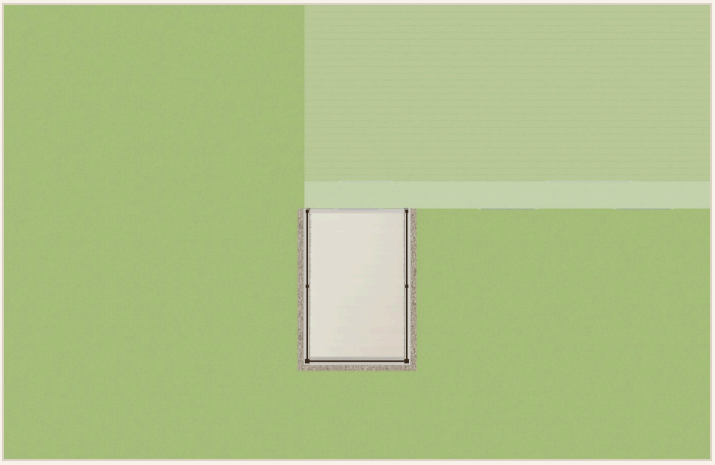
Isometric from the stair corner



Isometric from the open corner

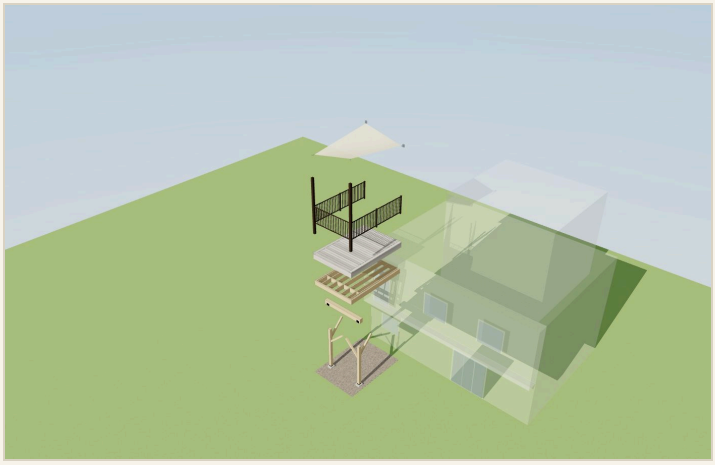


Isometric from under the frame

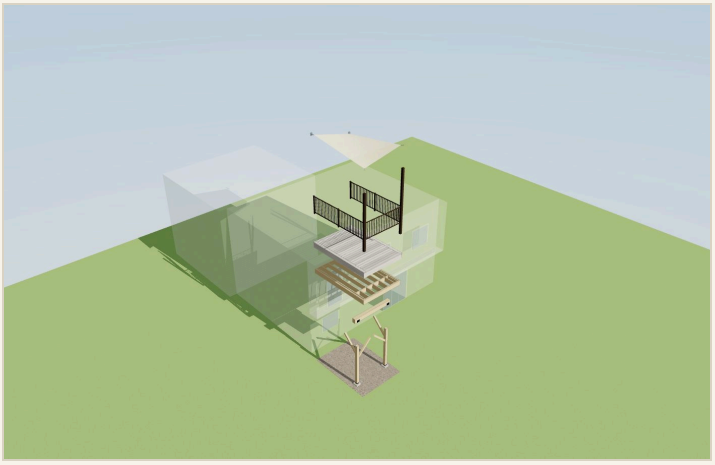


Plan from above

EXPLODED VIEWS every layer pulled apart



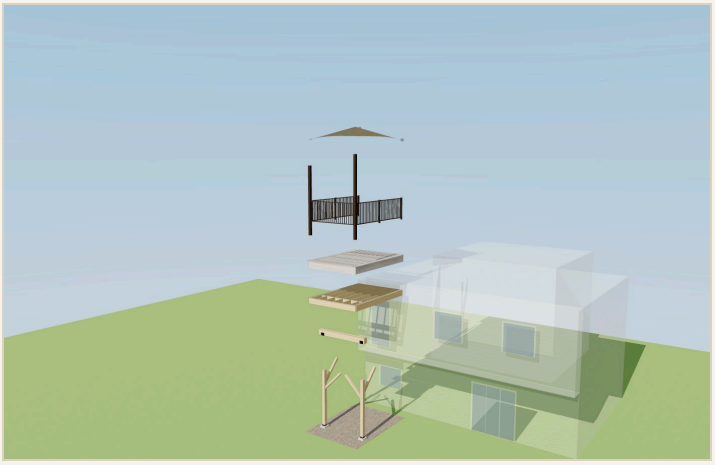
Exploded — every layer



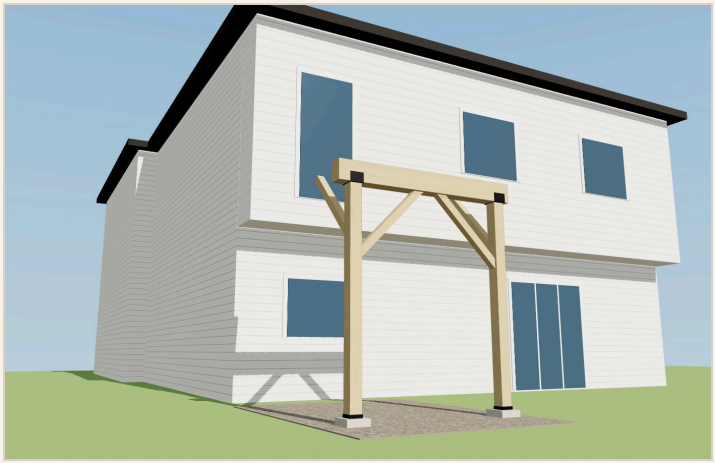
Exploded — from the open corner



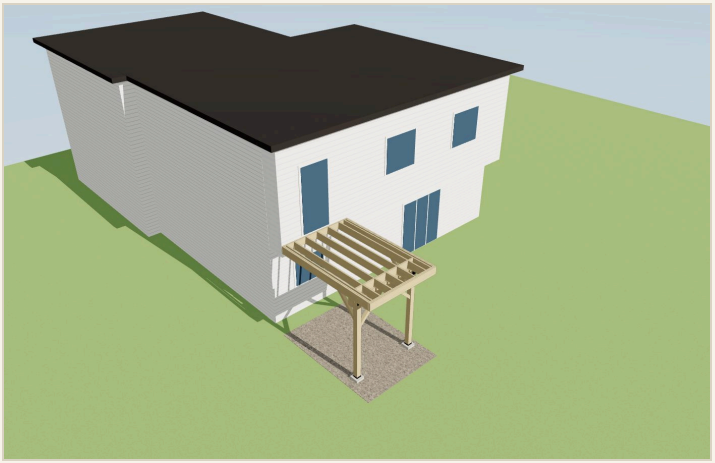
Exploded — from the front corner



Exploded — low angle



Posts and beams on the footings



Frame only — joists, rims, beams, ledger

INVESTMENT

PRICE

Total investment $23,842 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about $277/month

Check or ACH: $22,173 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Demolition & haul-off	$203
Foundation & structure	$5,537
Decking & fascia	$4,631
Railing	$3,543
Shade sail on 2 steel posts at the rail corners	$7,891
Site & project services	$368
Total, check or ACH	**$22,173**
Total, financed	**$23,842**

Engineering — $1,800 to $3,000. Stamped drawings by a Colorado engineer (30 psf snow). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

NEXT STEP

Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer and supplies the design package; you pay GSX the engineer's cost at no markup and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

3D MODEL



The interactive 3D model lives at the online version of this document.

GS Exterior Experts · Deck Division

Renders and drawings are design intent; dimensions on the sheets govern. Numbers hold 30 days.

GS EXTERIOR EXPERTS · DECK DIVISION · TAKEOFF + BUILD SET

5549 S Lansing Ct · Elevated Deck · Englewood

5549 S Lansing Ct, Englewood · Rachel Chiu · 8' x 12' deck, 10' above the yard, off the existing door · TrueFrame 2x10 @ 16" on a (2)4x10 Douglas fir beam and (2) 6x6 DF #1 posts on Diamond Piers · TimberTech Reserve Driftwood, single picture frame, CONCEALoc · Impression Rail Express, vertical balusters, modern top rail, Bronze · square shade sail on two steel posts at the rail corners

CHECK OR ACH	FINANCED	MONTHLY	SQUARE FEET	FOOTINGS	RAIL
$22,173	**$23,842**	**$277**	**92**	**2**	**31 LF**

DESIGN READS

Frame	7'-9" x 11'-5 1/2" outside to outside
Deck	7'-11 1/8" x 11'-6 9/16" over fascia · 92 SF · 10'-0" above grade
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 front and sides
Beam	DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50
Load	50 psf (40 psf live governs) · engineered
Decking	TimberTech Reserve Bronze — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

- nominal 8'-0" x 12'-0" -> frame 7'-9" x 11'-5 1/2": field boards cut to 7'-0", 24 full rows, deck over fascia 7'-11 1/8" x 11'-6 9/16"
- 2'-3 1/2" cantilever past the drop beam exceeds L/4 of the 9'-0 1/2" back-span — engineered condition (owner-directed)

SITE & CODE FLAGS internal

ENGINEER ENGINEERING DROP (2)4x10 DF#1: (2)4x10 not in IRC R507.5 — treated as (3)2x10; engineer to confirm IRC Table R507.5

ENGINEER ENGINEERING Shade sail: the sail posts stand on the deck as the rail corners — the sail's wind pull goes into the double rim, the frame, the knee-braced 6x6 posts and the house ledger:

		engineer checks the post-to-rim bolting, the frame and the lateral system for it ASCE 7 / IRC R507
ENGINEER	ENGINEERING	Shade sail: sail wind load (115 mph) on the 2 steel posts and the house anchors — engineer sizes the posts / footings and the anchor connection into the house framing (not into siding or fascia alone) ASCE 7 / IRC R301.2
ENGINEER	LEDGER	house floor cantilevers past the foundation — ledger attachment to a cantilevered floor needs an engineered detail or a freestanding deck IRC R507.9.1
ENGINEER	LEDGER	Arapahoe County deck guide: a ledger 'shall not be attached to a cantilevered floor' — attachment to a cantilever requires engineer approval. The 2' second-floor cantilever runs the full rear wall (assessor sketch): the stamped design either details the ledger into the cantilevered joists / blocking with the lateral ties back to the floor system, or makes the deck freestanding (a second beam on two posts at the house) AC-amended ICC Deck Guide (2021 IRC) / IRC R507.9.1
CODE	FOOTING	county guide: pier footings 12" x 36" minimum unless engineered — submit the Diamond Pier DP-50 evaluation report as the engineered alternate (frost 36") and have the engineer list it on the stamped set Arapahoe deck permit guide (rev. 8/20/2024) / IRC R403.1.4
CODE	RAIL	guards above 30": 36" min., openings under 4", 200 lb point load + 50 psf infill — submit the TimberTech Impression Rail Express code report / ESR IRC R312 / Table R301.5
CODE	SUBMITTAL	county submittal: site plan with lot / block and distances to every line, scaled drawings with every span, member, hanger, beam build-up and post anchorage, ledger fastener type / size / spacing, the completed county-amended ICC deck guide (page 3), and the engineer's stamp (outside the IRC tables: cantilever attachment, 2-post (2)4x10) Arapahoe Residential Deck Permit guide (rev. 8/20/2024)
CHECK	SETBACK	easement on record: Check plat R1535678 for rear / side utility and drainage easements — no footings or structure inside it plat
CHECK	SITE	shade sail: check whether the county wants a separate permit for the sail structure; the sail must meet the setbacks county
CHECK	FOOTING	Diamond Pier: pins 50" — locate utilities (811) before driving; bearing 2,214 lb/post vs 3,300 lb allowable (manufacturer chart, confirm)
CHECK	LEDGER	verify the rim board behind the ledger is solid 2x/LVL/engineered rim (not I-joist web or a hollow band) before the ledger fasteners go in IRC R507.9.1.1
CHECK	SETBACK	R-P PUD Cherry Creek Vista: rear setback 20' for single-family detached, 10' under the Units 6 & 13 standard (the lot is 6,401 SF, under the 7,000 SF SFD minimum). GIS puts the new deck ~13' off the west line — passes at 10', fails at 20'. Decks get no projection allowance (LDC). Zoning Division to confirm the column and which line is the rear (corner lot) before submittal; pull P75-028 and plat R1535678 2nd Amended PDP P75-027 (1975) / Arapahoe LDC

CHECK	SETBACK	corner lot: E Berry Ave street side needs 10' — deck ~23'-8" clear; south interior side 5' — deck ~33' clear PDP P75-027
CHECK	SITE	HOA: Cherry Creek Vista II HOA (PO Box 3361, Englewood 80155, 303-770-1469) — confirm whether architectural approval is needed county GIS
CHECK	SITE	811 locate before the piers · FEMA zone X · not in the county's draft wildfire-resiliency area (no WUI requirements) · South Metro Fire Rescue Colorado 811 / FEMA / Arapahoe GIS
INFO	SETBACK	HOA — architectural approval before start (color, rail, height)
INFO	ENGINEERING	posts 8'-0 1/16": 4x4 knee braces both directions at every post are in the takeoff — engineer confirms the lateral design DCA6 / IRC R507.4
INFO	LEDGER	lateral: 4 x DTT1Z (750 lb each) into house floor framing IRC R507.9.2
INFO	WIND	basic wind speed 115 mph — standard connectors
INFO	SITE	keep any basement egress window under or near the deck clear (county guide) Arapahoe deck permit guide

THE ORDER Decks & Docks · GSEXT1 · ship to the job

Lumber

NET	ORDER	UNIT	ITEM	WHY
15	**16**	ea	2x10x12 #1 True Frame Joist SYP GC	+1 cull (5 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · 2 blocking, front rim ply · knee-brace block, ledger, rail post block · rail post block)
2	**2**	ea	4x10x8 Douglas Fir #1	exact (2 drop beam: DROP (2)4x10 DF#1)
2	**2**	ea	6x6x10 Douglas Fir #1	exact (yields 2 posts at ~8'-0 1/16")
2	**3**	ea	4x4x8 Douglas Fir #1 — knee braces	+1 (4 braces at 3'-9 15/16", 45° plumb cuts, 2'-6" legs: one along the beam, one up to the joists at every post)

Footings

NET	ORDER	UNIT	ITEM	WHY
2	**2**	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock
2	**2**	ea	Simpson ABA66Z adjustable post base 6x6 on the pier bolt	exact

Hardware

NET	ORDER	UNIT	ITEM	WHY
4	**5**	ea	1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)	+1
2	**3**	ea	1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)	+1

NET	ORDER	UNIT	ITEM	WHY
2	**3**	ea	1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)	+1
14	**14**	ea	Simpson LUS28Z face-mount hanger 2x8/2x10	exact (7 joists x 2 ends)
4	**4**	ea	Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)	exact (2 side rims x 2 ends) — confirm stock
1	**1**	box	Camo 9ga HDG hanger nail 3" 5 lb (~350)	84 needed of ~350
1	**1**	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	146 needed of ~650
1	**1**	box	FastenMaster LedgerLOK 3-5/8" 50 ct	9 needed
4	**4**	ea	Simpson DTT1Z lateral tie kit w/ SDS screws	exact (2 near each end of every ledger, into house floor framing)
9	**9**	ea	Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)	exact (7 joists + 2 rims at each drop beam)
2	**2**	ea	Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal	exact
10	**12**	ea	1/2" x 8" HDG carriage bolt + nut + washer	+2 ((2)4x beam plies through-bolted, 2 rows staggered @ 24" OC, 2" from top and bottom)
1	**1**	box	Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct	24 needed (bases 24 · caps 0)
1	**1**	box	GRK RSS #10 x 3-1/8" structural screw 100 ct	94 needed (laminations 70 · blocking 24)
1	**1**	box	FastenMaster HeadLOK 6" (50 ct) — rail posts 4 ea	24 needed (6 rail posts x 4)

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	**1**	set	Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams	exact
2	**2**	roll	Nichigo G-Tape 3035 2" x 65' black	93' needed of 130'
1	**1**	roll	Nichigo G-Tape 3035 4" x 65' black	38' needed of 65'

Decking

NET	ORDER	UNIT	ITEM	WHY
12	**14**	ea	TimberTech Reserve Driftwood 1x6x16 Grooved	+2 (A: 24 rows @ 7'-0", 2 per 16' board)
1	**1**	ea	TimberTech Reserve Driftwood 1x6x12 Square Edge — borders	exact (right border 11'-6 9/16")
1	**2**	ea	TimberTech Reserve Driftwood 1x6x20 Square Edge — borders	+1 (left border 11'-6 9/16" + front border 7'-11 1/8")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)	168 needed · 500 supplied

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct	106 screws needed (first row 7 · border 58 · fascia 41)

Fascia

NET	ORDER	UNIT	ITEM	WHY
3	**4**	ea	TimberTech Reserve Fascia Driftwood 0.58x11.95x12	+1 (left fascia 11'-6 1/16" · right fascia 11'-6 1/16" · front fascia 7'-10 3/16")

Rail

NET	ORDER	UNIT	ITEM	WHY
4	**4**	ea	TimberTech Impression Rail Express 6' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze	exact (cut to 5'-3 3/4")
1	**1**	ea	TimberTech Impression Rail Express 8' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze	exact (cut to 7'-1 1/2")
4	**4**	kit	TimberTech Impression Rail Express 2" aluminum post kit 36" (post, base, cap, skirt), Bronze	exact (2 CORNER, 2 END, 2 LINE; 2 corners are the sail posts)
1	**1**	ea	TimberTech Bronze touch-up paint (cut ends)	exact

Shade sail

NET	ORDER	UNIT	ITEM	WHY
73	**73**	SF	Shade sail — custom commercial-grade HDPE shade sail, 4 corners, hypar twist, Sand, stainless D-rings, sewn-in perimeter cable	made to the measured attachment points, ~6% under for tensioning
1	**1**	ea	Steel sail post 4" x 4" x 1/4" HSS, galvanized + powder coat Bronze, pad eye welded at the top, rail-bracket tabs — 11' (9'-1 3/16" above the deck + 1'-7" down the rim)	post 1: the rail corner post at this corner
1	**1**	ea	Steel sail post 4" x 4" x 1/4" HSS, galvanized + powder coat Bronze, pad eye welded at the top, rail-bracket tabs — 10' (7'-7 3/16" above the deck + 1'-7" down the rim)	post 2: the rail corner post at this corner
6	**8**	ea	5/8" x 10" HDG through-bolt + nut + plate washers — sail posts through the double rim and blocking	+2 (3 per post)
1	**1**	ea	2x10x8 #1 True Frame — solid blocking behind each sail post (rim to PF joist and the next joist)	4 blocks
1	**1**	kit	Sail hardware: 4 stainless turnbuckles (jaw-jaw), D-shackles, snap hooks, 2 house anchor plates with through-bolts / structural screws into the framing	316 stainless, marine grade

Site

NET	ORDER	UNIT	ITEM	WHY
64	**64**	SF	Demo existing deck + haul-off	labor line — field verify the existing deck

FASTENER & CONNECTOR SCHEDULE

Knee braces	4x4 DF #1 at 45°, 2'-6" legs, both directions at every post: one along the beam toward mid-span, one toward the house up to a 2x10 block between the joists; (1) 1/2" through-bolt each end, 2" min. edge distance
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Beam	(2)4x plies through-bolted with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a bolt pair 6" from each end; CCQ cap with SDS screws at each post
Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	CONCEALoc at every joist in every gap (168)
Face screws	Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
Rail	IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists	2x10	11'-1"	5	@ 16" OC, crown up
Picture-frame joists	2x10	11'-1"	2	centre 5 1/4" from each frame face
Side rim plies	2x10	11'-1"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies	2x10	7'-9"	2	inner ply takes the hangers; outer ply laminated after
Ledger	2x10	7'-9"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking	2x10	1'-2 1/2"	6	1 row at 9'-2" from the wall
Field boards run A	Reserve Driftwood	7'-0"	24	boards run parallel to the house
DROP (2)4x10 DF#1 beam	4x10	7'-8"	2	y 9'-2"; posts at 9" / 7'-3"
Posts under DROP (2)4x10 DF#1	6x6	8'-0 1/16"	2	field-measure each
Knee braces	4x4	3'-9 15/16"	4	45° both ends, 2'-6" legs · along the beam toward mid-span + up to a block between the joists
Left border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist
Right border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist

MEMBER	STOCK	LENGTH	QTY	NOTE
Front border	Reserve Driftwood square	7'-11 1/8"	1	mitre at the corners; dividers on the doubled joist
Left fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia	fascia	7'-10 3/16"	1	flush under the board nose; full board at the front corners, short piece at the house

THE STACK internal

Materials (raw)	$8,035.22
Materials × 1.15	$9,240.50
Sales tax 4.25% on factored materials	$392.72
Labor (rate card)	$4,104.00
Work — carries the margin	$13,737.22
General conditions at cost	$368.00
Cost	$14,105.22
Sell — check or ACH at 37.0% GM (blended — 31 LF rail at $114.29/LF ($3,543), everything else at 37.0% GM)	**$22,173** · $241.01/SF
Sell at the 40% floor (the deck at 40%, rail at its rate)	$23,086
Financed	**$23,842** · $277/mo at 6.99% × 10 yr
Gross profit	$8,068
Engineering — outside the price, at cost	$1,800–$3,000

Labor

LINE	QTY	UNIT	RATE	EXT
Demolition — existing deck	64	SF	$2.00	$128.00
Diamond Pier install (replaces caisson)	2	pier	$50.00	$100.00
Frame	92	SF	$10.00	$920.00
Decking & fascia	92	SF	$8.00	$736.00
Railing	31	LF	$20.00	$620.00
Shade sail — 2 steel posts bolted through the rim as the rail corners, blocking, house anchors, hang and tension	1	lot	$1,600.00	$1,600.00

General conditions — at cost, no margin

ITEM	AMOUNT	WHY
General conditions	$368	$4/SF x 92 SF

What the price includes — sell values

ITEM	VALUE
Demolition & haul-off	$203
Foundation & structure	$5,537
Decking & fascia	$4,631
Railing	$3,543
Shade sail on 2 steel posts at the rail corners	$7,891
Site & project services	$368
Total, check or ACH	**$22,173**
Total, financed	**$23,842**

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

5549 S LANSING CT — ELEVATED DECK, 8' X 12' AT 10' (TIMBERTECH RESERVE DRIFTWOOD, IRX BRONZE)

2x10 SYP frame at 16", (2)4x10 beam, 6x6 posts — on 2 Diamond Pier footings. TimberTech Reserve Driftwood decking, Impression Rail Express, vertical balusters, modern top rail rail.

92 SF · 1 zone · **10'-0"** above grade, engineered · **30 psf** snow design

Prepared for Rachel Chiu · 5549 S Lansing Ct, Englewood

PRICE

Total investment $23,842 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about $277/month

Check or ACH: $22,173 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Demolition & haul-off	$203
Foundation & structure	$5,537
Decking & fascia	$4,631
Railing	$3,543
Shade sail on 2 steel posts at the rail corners	$7,891
Site & project services	$368
Total, check or ACH	**$22,173**
Total, financed	**$23,842**

Engineering — $1,800 to $3,000. Stamped drawings by a Colorado engineer (30 psf snow). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

INCLUDED — WHAT IT IS

ELEMENT	SELECTION
Decking	TimberTech Composite Reserve, Driftwood. Grooved boards, 3/16" gaps, matching fascia. Class C, not WUI; 30-yr product / 30-yr fade & stain.
Railing	TimberTech Impression Rail Express, vertical balusters, modern top rail, Bronze, 36". 31 LF.
Frame	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"; 2-ply 2x10 front and sides; DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb).
Foundation	2 Diamond Pier foundations (DP-50/50) — no excavation.

ELEMENT	SELECTION
Design load	30 psf snow, stamped by a Colorado engineer.
Hardware	Simpson ZMAX galvanized connectors. CONCEALoc fasteners.
Site	The existing deck removed and hauled off.

PLAN

7'-9" x 11'-5 1/2" outside to outside · 92 SF · 10'-0" above grade · 31 LF rail

SCHEDULE

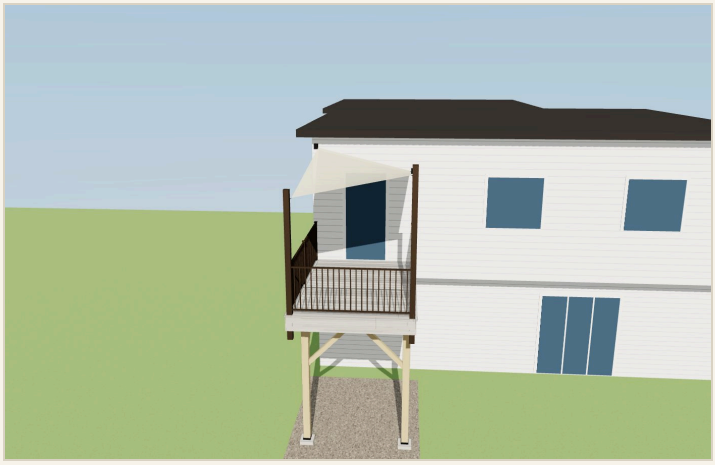
1. Engineering & ordering (2–3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811), old deck out.
3. Day 2 — layout, 2 Diamond Piers driven, ledger set.
4. Days 3–4 — posts and beams set.
5. Days 5–6 — frame, flashing at every wall.
6. Days 7–8 — decking, borders first.
7. Days 9–9 — IRX Baluster rail.
8. Day 10 — final clean; end of construction.

NEXT STEP

Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer and supplies the design package; you pay GSX the engineer's cost at no markup and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

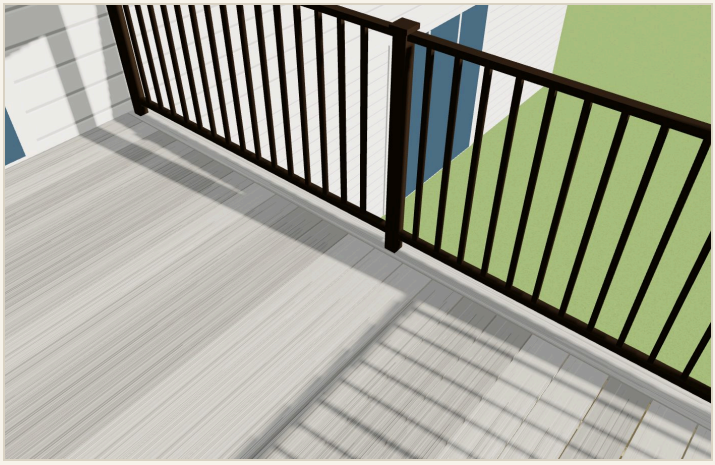
RENDERS 3D model · design intent · frame shown as specified



From the yard



From the front corner



Standing on the deck



Under the deck — posts, beams, footings



Jul 15 · 9 AM — sun E, 35° up



Jul 15 · 12 PM — sun SE, 67° up



Jul 15 · 3 PM — sun WSW, 60° up

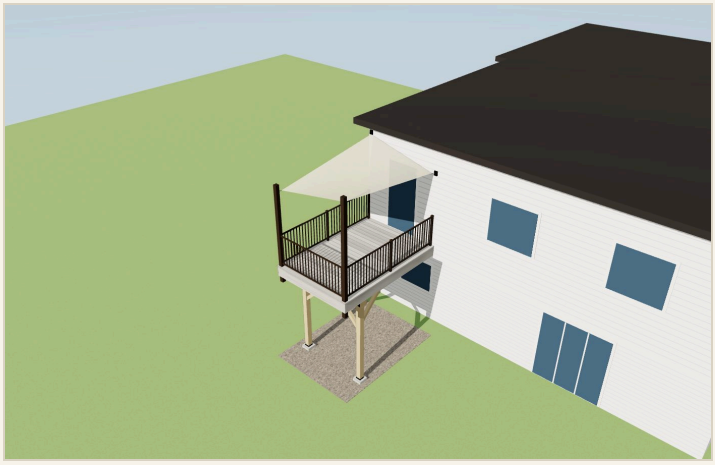


Jul 15 · 6 PM — sun W, 26° up

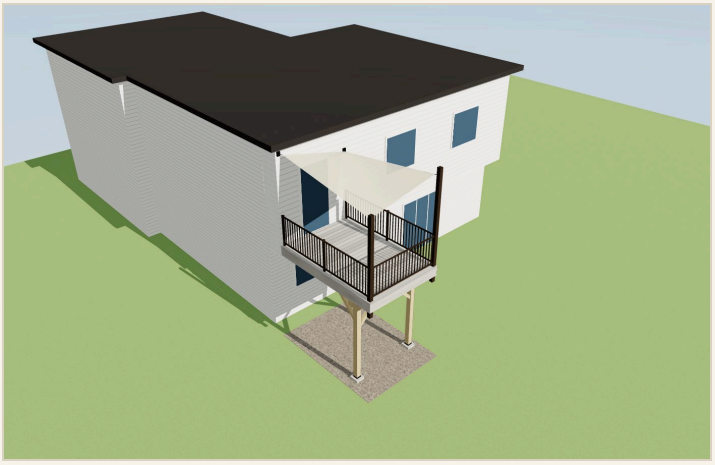


Dec 21 · 3 PM — sun SW, 14° up

ISOMETRICS



Isometric from the stair corner



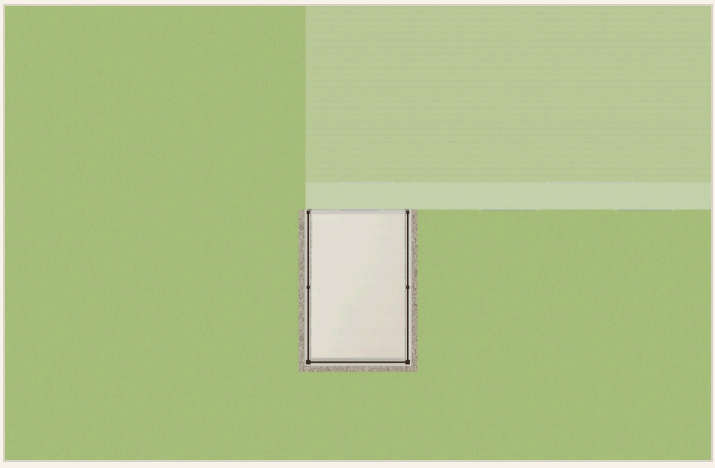
Isometric from the open corner



Low isometric

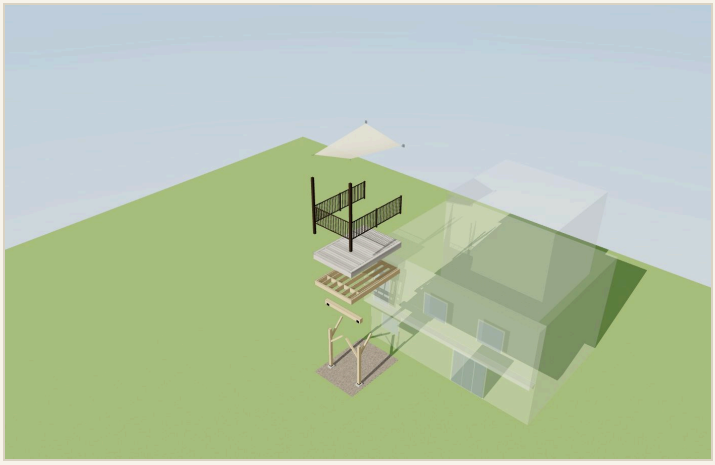


Isometric from under the frame

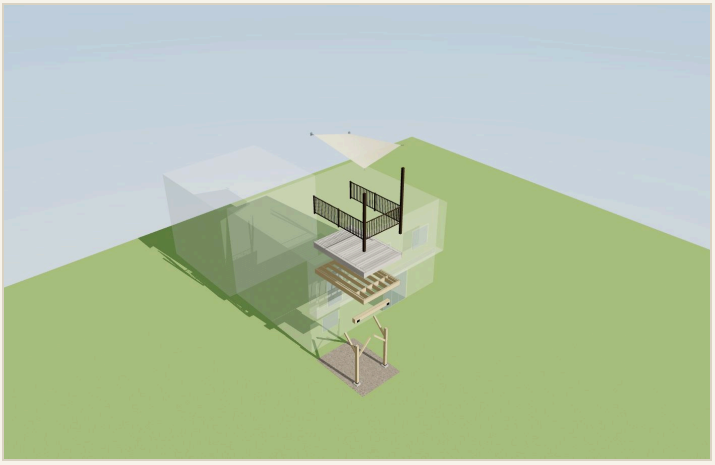


Plan from above

EXPLODED



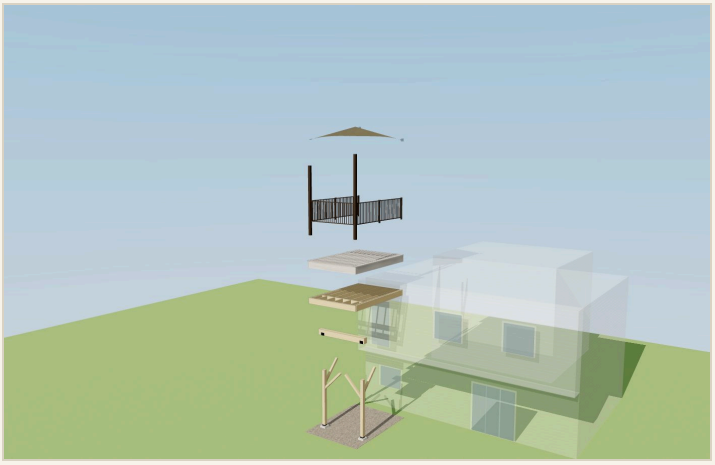
Exploded — every layer



Exploded — from the open corner



Exploded — from the front corner



Exploded — low angle

G-001 General notes · design criteria

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PROJECT

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech Reserve Driftwood, IRX bronze)
5549 S Lansing Ct, Englewood CO 80111
Prepared for Rachel Chiu
92 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; 30 psf ground snow, roof 30 psf non-reducible (Arapahoe County design criteria); deck tables at 40 psf live + 10 psf dead
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult (3-second gust), Exposure B (suburban), Risk Cat. II — county design-criteria table (2015 cycle; confirm the 2021 table)
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC / IBC as adopted by Arapahoe County (Res. 21-394, eff. 4/1/2022); IRC R507 decks with the county-amended Colorado ICC Deck Building Guide; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces
Beams	DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 front and sides
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Decking	TimberTech Reserve Bronze — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

SHEET INDEX

G-001	General notes · design criteria
C-100	Site plan
A-001	Isometric · renders
A-002	Isometric — assembled, labeled
A-003	Isometric — exploded, labeled
S-100	Foundation plan
S-101	Framing plan
S-102	Structure isometric, labeled
A-101	Decking plan
A-201	Railing plan + elevation
A-202	Elevations
A-301	Section + front elevation
D-501	Details
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)

SITE

Rear / front lot line: deck edge 13.0' from the line, setback per the PUD...
Side lot line: deck edge 23.7' from the line, 5' required
Easement: Check plat R1535678 for rear / side utility and drainage...
Unincorporated Arapahoe County (Cherry Creek CDP; Englewood mailing address). Parcel 2075-14-4-05-001 / PIN 031772427, Lot 41 Blk 9 Cherry Creek Vista 6th Flg Amend Resub (plat R1535678), 6,401 SF corner lot (front S Lansing Ct east, street side E Berry Ave north). R-P PUD Cherry Creek Vista (Z75-030 / P75-027 / P75-028). Deck on the WEST (rear) wall, north end, hung on the 2' second-floor cantilever that runs the full 33' wall (assessor sketch). Tax: CO 2.9 + RTD 1.0 + CD 0.1 + Arapahoe 0.25 = 4.25% (no city).

FLAGS (INTERNAL — remove before the permit submittal)

ENGINEER	DROP (2)4x10 DF#1: (2)4x10 not in IRC R507.5 — treated as...
ENGINEER	Shade sail: the sail posts stand on the deck as the rail corners...
ENGINEER	Shade sail: sail wind load (115 mph) on the 2 steel posts and the...
ENGINEER	house floor cantilevers past the foundation — ledger attachment...
ENGINEER	Arapahoe County deck guide: a ledger 'shall not be attached to a...
CODE	county guide: pier footings 12" x 36" minimum unless engineered —...
CODE	guards above 30": 36" min., openings under 4", 200 lb point load...
CODE	county submittal: site plan with lot / block and distances to...

No scale

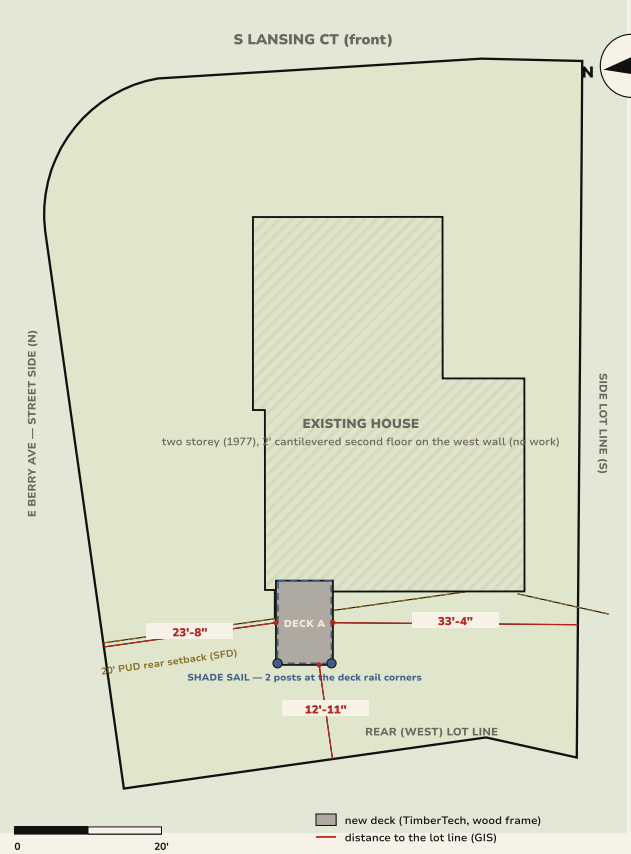
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

C-100 SITE PLAN

Parcel, existing house, the new deck, distances to the lot lines, setbacks

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PARCEL

5549 S Lansing Ct, Englewood CO 80111
Parcel 2075-14-4-05-001 · PIN 031772427
Lot 41, Block 9, Cherry Creek Vista 6th Filing Amended Resub (plat R1535678)
Unincorporated Arapahoe County · zoning R-P PUD (Z75-030 / P75-027 / P75-028)
Lot 6,401 SF (GIS) · corner lot · flood zone X
House 1977, 2 storey, 1,704 SF + 660 SF basement + 441 SF garage (assessor)
HOA: Cherry Creek Vista II

LOT COVERAGE

Lot area	6,401 SF
Lot coverage limit	none in the PUD plan
Existing deck (assessor)	48 SF (8' x 6') — removed
New deck	8' x 12' · ~92 SF of frame, 10' above grade

SETBACKS

Elevated decks get no setback projection allowance (Arapahoe LDC) — the deck must meet the setbacks
GIS puts the new deck front ~13' off the west line: passes the 10' standard, not the 20' — confirm the governing column and the rear line (corner lot) with Zoning before submittal
E Berry Ave street side 10' required — ~23'-8" clear; south side 5' — ~33' clear
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the existing 8' x 6' deck and its two posts / pads; haul off
· Two Diamond Pier DP-50 foundations (no excavation spoils), 6x6 DF #1 posts with 4x4 knee braces
· New ledger on the 2' cantilevered second floor per the engineer's detail; membrane + Z-flash, (4) DTT1Z lateral ties
· No stairs — the deck is reached from the house door only
· Call 811 before the piers; protect the lawn and the patio slab at the north wall

Arapahoe County GIS (address point, parcel, 2024 roof prints, R-P zoning, jurisdiction, flood, wildfire layers), county assessor record and sketch, 2nd Amended PDP P75-027 setback table, county deck permit guide (rev. 8/20/2024) and the county-amended ICC deck guide. GIS lines are not a survey — an ILC governs if Zoning asks.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

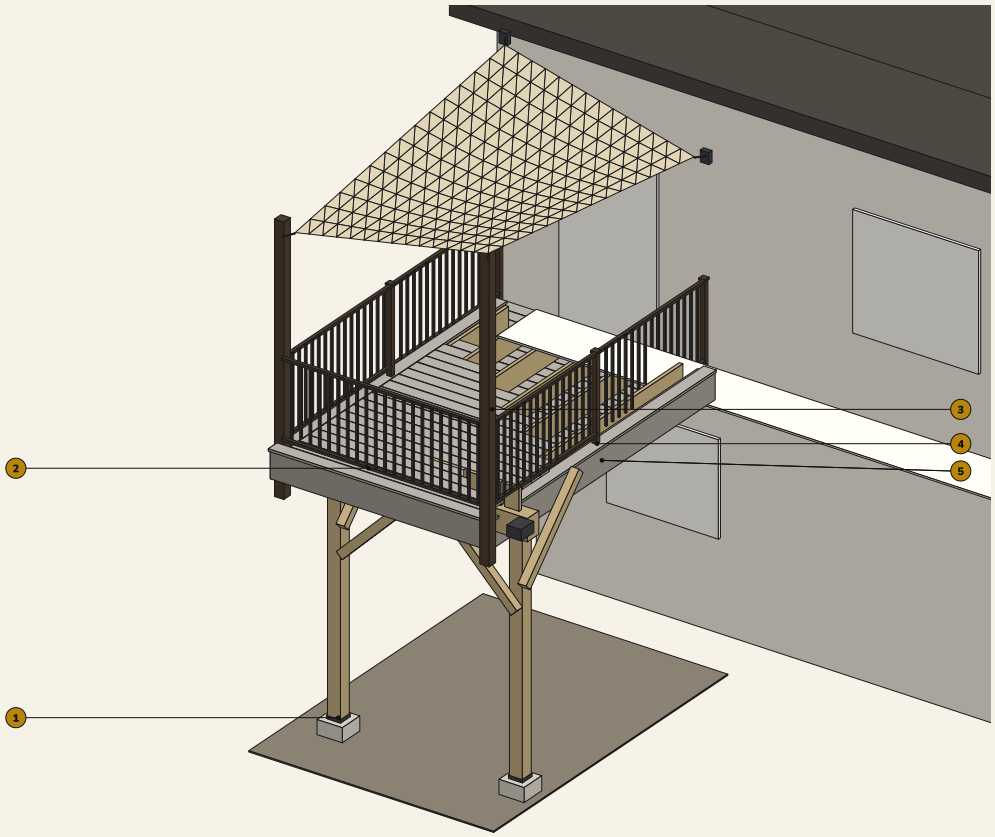
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
3. TimberTech Reserve Driftwood field decking, hidden fasteners
4. Picture frame + breaker boards — TimberTech Reserve Driftwood
5. Fascia

House shown for context only (the deck work only). The frame under the decking is called out on S-102 and A-003.

Axonometric · not to scale · the model the plans are drawn from

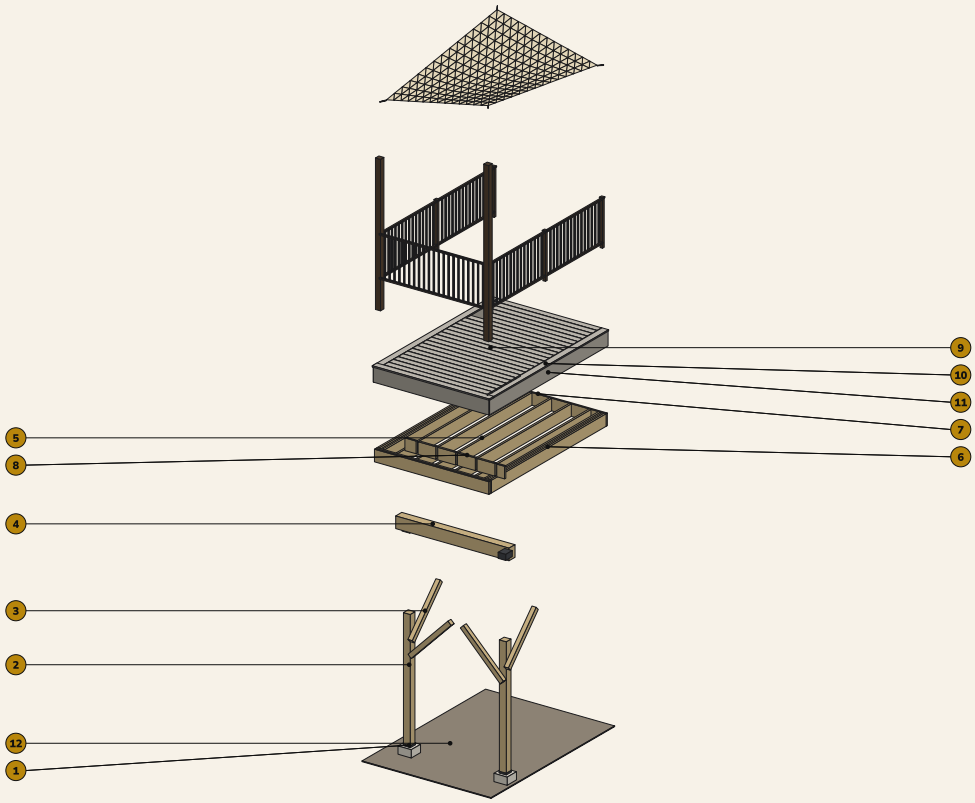
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-003 ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts + knee braces · beams · frame · decking · rail

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 FOR REVIEW



KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. TimberTech Reserve Driftwood field decking, hidden fasteners
10. Picture frame + breaker boards — TimberTech Reserve Driftwood
11. Fascia
12. Gravel over 6-mil poly under the decks

Layers lifted in build order. Lifts are for clarity only; every member returns straight down to its place.

Axonometric · not to scale · the model the plans are drawn from

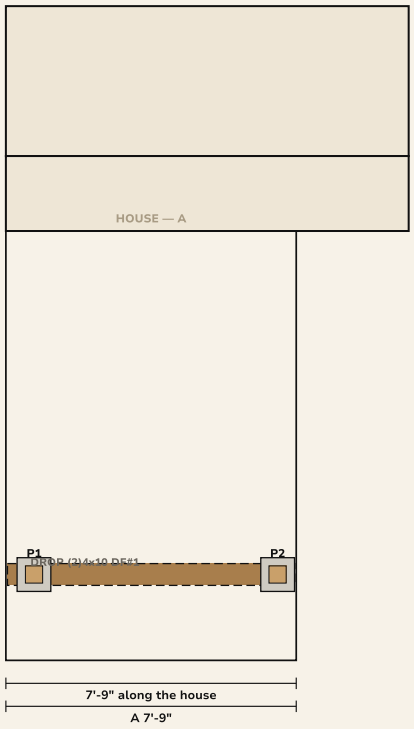
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	9"	9'-2"	A	2,214 lb
P2	7'-3"	9'-2"	A	2,214 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 2,214 lb

Plan · not to scale · house at the top · dimensions govern

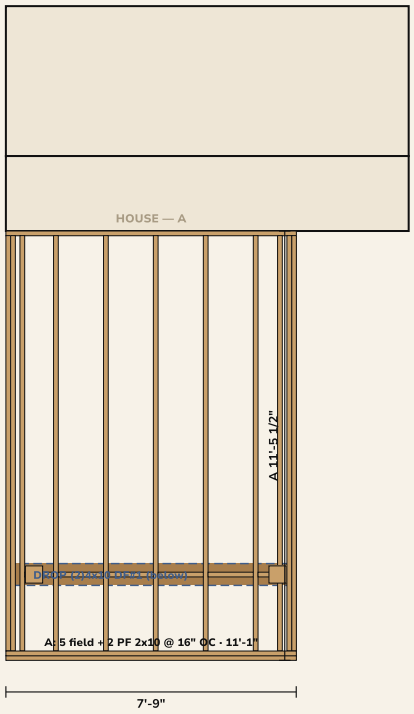
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRAMING NOTES

7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"

2-ply 2x10 front and sides

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

blocking one row over the beam, staggered

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Connectors:

1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post): 4

1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam): 2

1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block): 2

Simpson LUS28Z face-mount hanger 2x8/2x10: 14

Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 4

Simpson DTT1Z lateral tie kit w/ SDS screws: 4

Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden): 9

Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal:

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

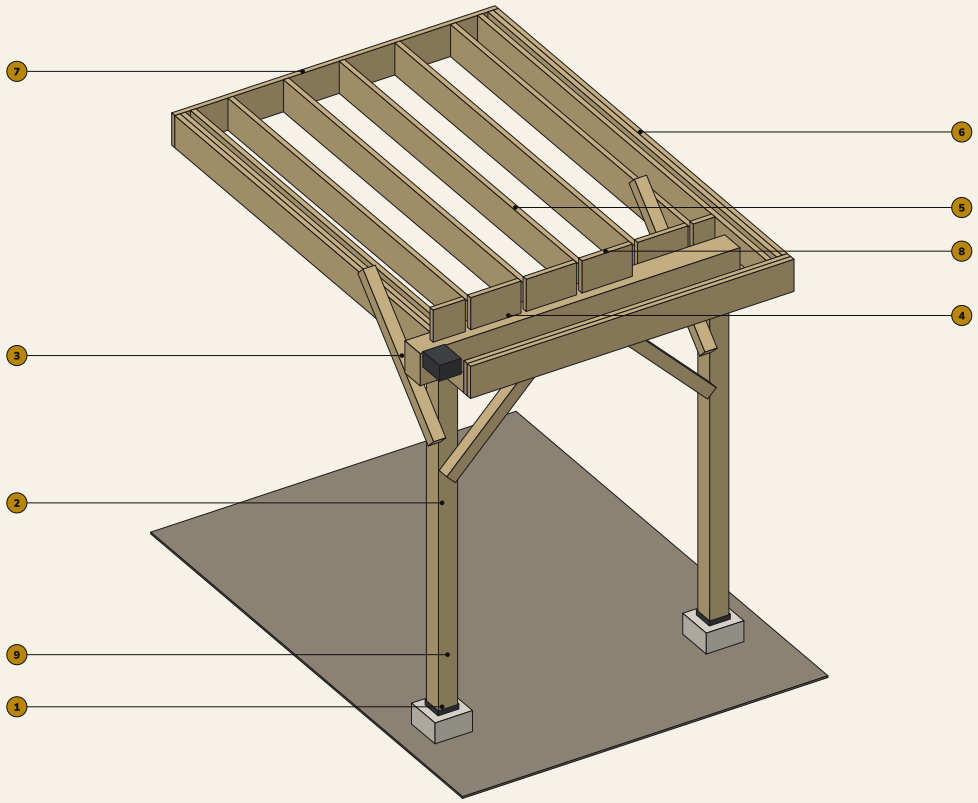
S-102 Structure isometric, labeled

S-102 **STRUCTURE ISOMETRIC — FRAME, LABELED**

Piers, post bases, posts, knee braces, post caps, beam, joists, double rims, ledger, blocking

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. Gravel over 6-mil poly under the decks

Decking, rail and house removed to show the frame. Joists hang from the ledger in face-mount hangers, bear on the drop beam with H2.5AZ ties; double rims in concealed-flange hangers.

Axonometric · not to scale · the model the plans are drawn from

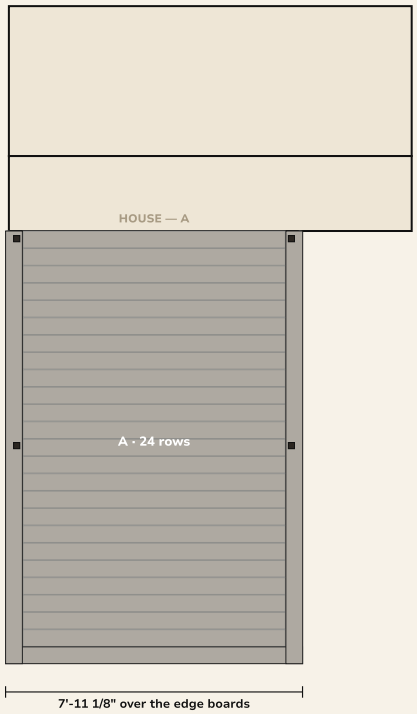
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-101 DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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DECKING

Field: TimberTech Reserve Driftwood, 5.36" x 0.94", 3/16" gaps

Border: Reserve Driftwood

Boards run parallel to the house; edge overhang 1 1/16" over the fascia

CONCEALoc at every joist in every gap (168)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per zone: A 24 (no rips)

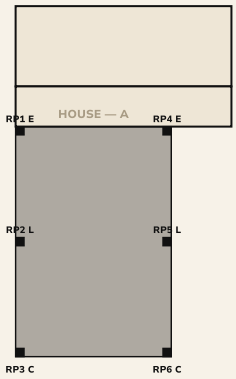
A-201 Railing plan + elevation

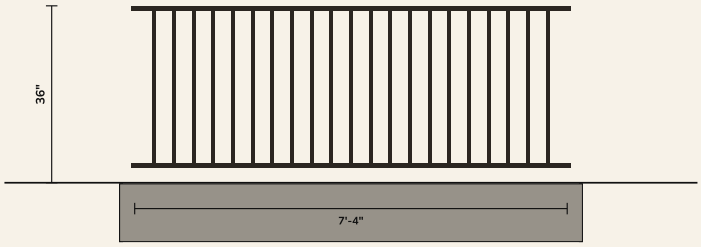
A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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RAILING

IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

Bays: left 5'-6 1/4" · left 5'-6 1/4" · right 5'-6 1/4" · right 5'-6 1/4" · front 7'-4"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.

Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

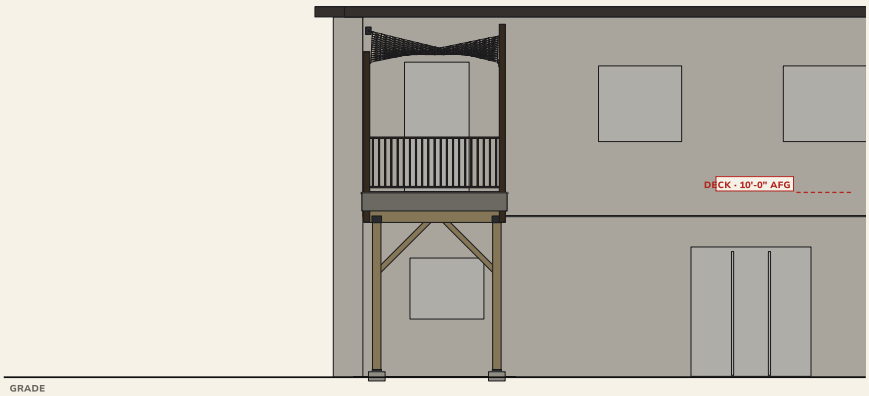
A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

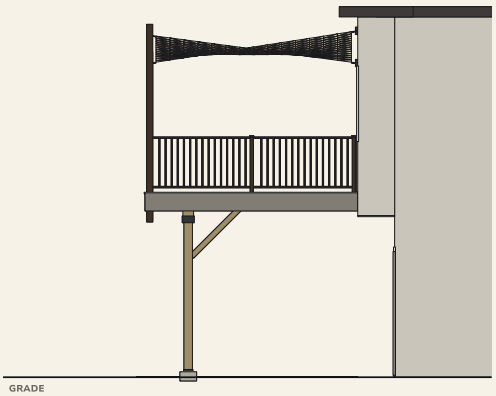
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRONT ELEVATION — from the yard



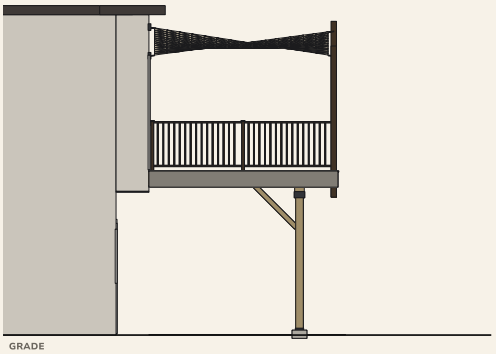
RIGHT END ELEVATION



NOTES

· Deck surface 10'-0" above grade
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Deck frame: 2x10 #1 True Frame GC @ 16" OC on a (2)4x10 DF #1 drop beam, 2 posts with 4x4 knee braces both ways
· Guard 36" min. on every open side — the surface is 10'-0" above grade (IRC R312.1)

LEFT END ELEVATION



A-301 Section + front elevation

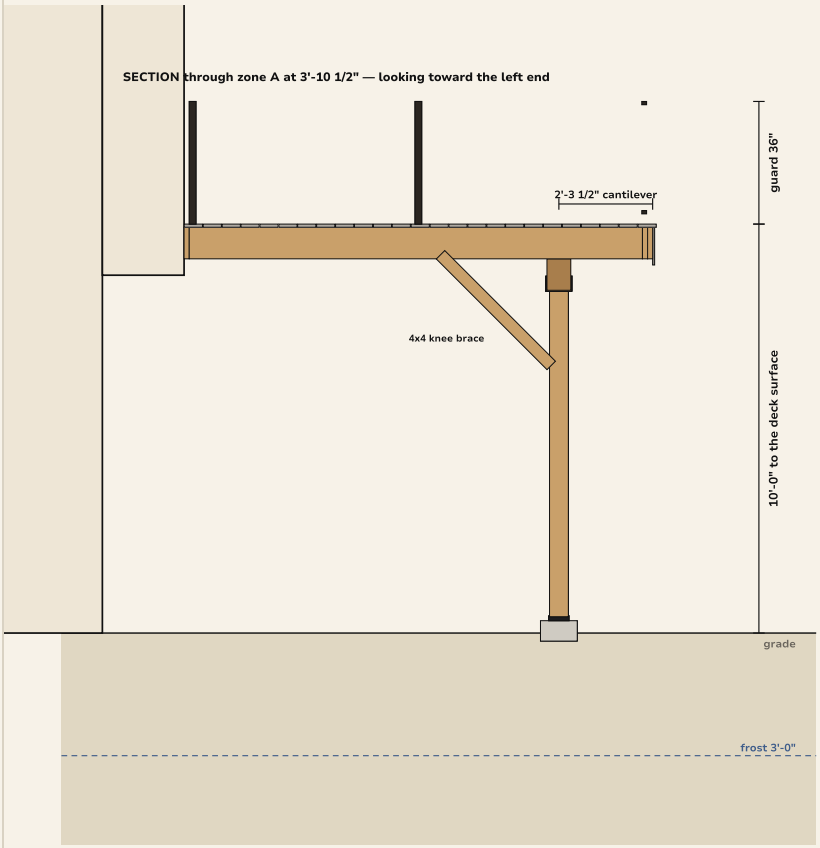
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

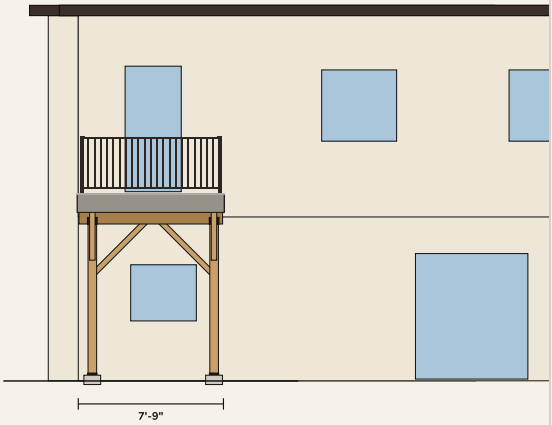
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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SECTION through zone A at 3'-10 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Posts 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

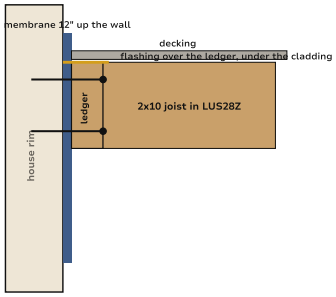
D-501 DETAILS

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

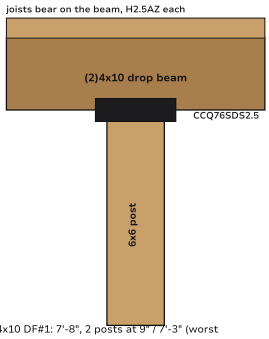
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1 LEDGER AT THE HOUSE



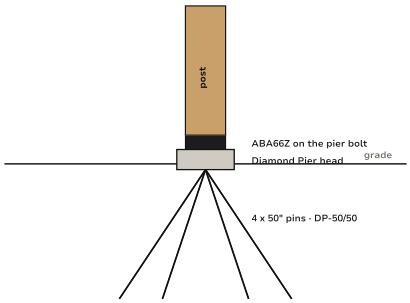
2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

2 POST · BEAM · CAP

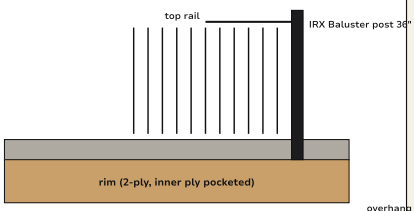


DROP (2)4x10 DF#1: 7'-8", 2 posts at 9' 7 7'-3" (worst 2,214 lb)

3 FOOTING · POST BASE

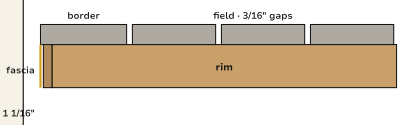


4 RAIL POST · GUARD



IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

5 DECK EDGE · BORDER



CONCEALoc at every joist in every gap (168)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.

Schematic details, not to scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

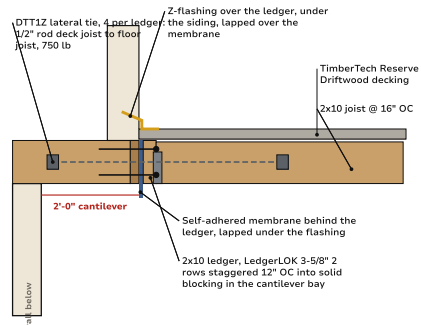
D-502 CONNECTION DETAILS — FRAME

Ledger on the cantilever · hangers · joists on the beam · (2)4x10 on the cap · pier + base · knee braces

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

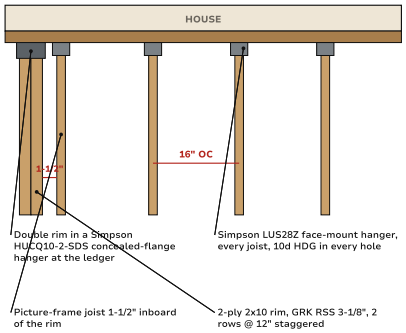
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1 LEDGER ON THE CANTILEVERED FLOOR — SECTION

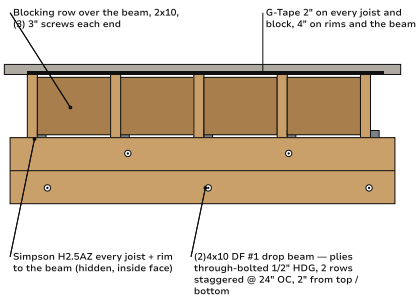


ENGINEER: Arapahoe County bars a ledger on a cantilevered floor without engineer approval. The stamped detail governs: ledger into the blocked cantilever joists, or a freestanding beam on posts at the house.

2 HANGERS — PLAN AT THE LEDGER + SIDE RIM



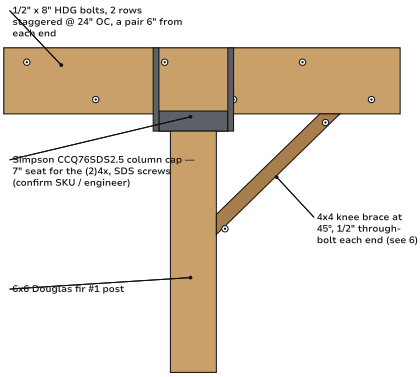
3 JOISTS ON THE DROP BEAM — SECTION



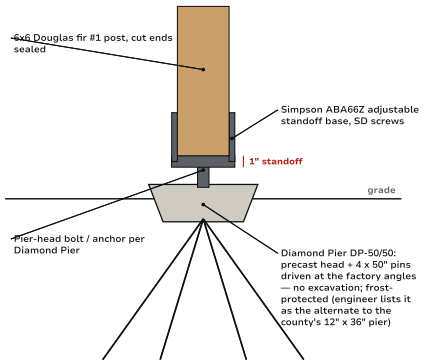
Joists bear the full beam width

Cantilever 2'-3 1/2" past the beam — over L/4: engineer

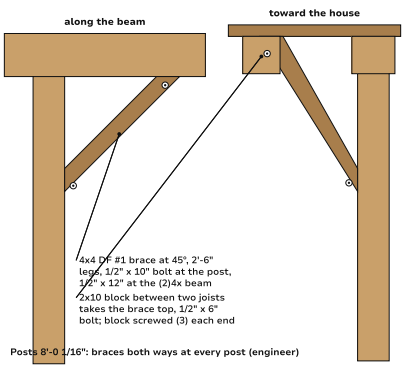
4 (2)4x10 BEAM ON THE 6x6 — ELEVATION



5 POST BASE ON THE DIAMOND PIER — SECTION



6 KNEE BRACES — BOTH DIRECTIONS



Posts 8'-0 1/16": braces both ways at every post (engineer)

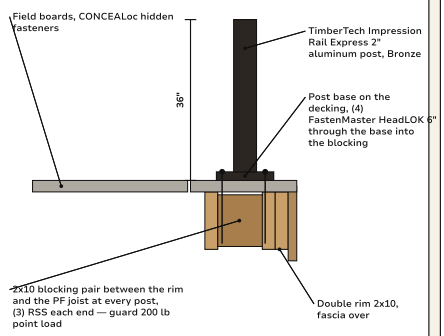
D-504 CONNECTION DETAILS — DECK EDGE, RAIL, LATERAL

Rail post mounting · picture frame + fascia · guard bay · DTT1Z lateral ties · joist tape · fastening schedule

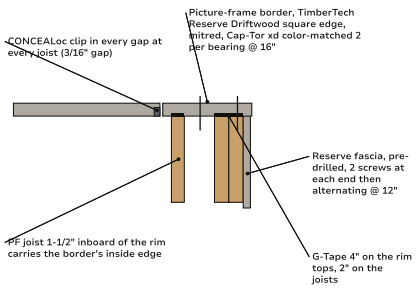
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 FOR REVIEW

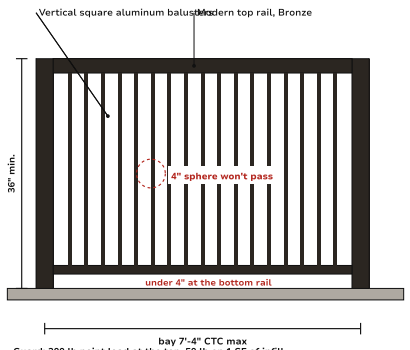
1 RAIL POST — SURFACE MOUNT ON BLOCKING



2 DECK EDGE — PICTURE FRAME + FASCIA

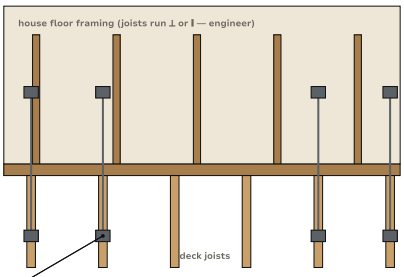


3 GUARD — ONE BAY, ELEVATION



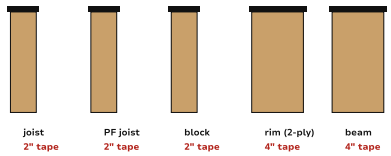
Guard: 200 lb point load at the top, 50 lb on 1 SF of infill (IRC Table R301.5) — submit the rail code report

4 LATERAL LOAD — DTT1Z TIES (PLAN)



(4) Simpson DTT1Z, 2 near each end of the ledger: 1/2" threaded rod through the ledger and the band into a floor joist (or blocking) inside — 750 lb each (IRC R507.9.2)

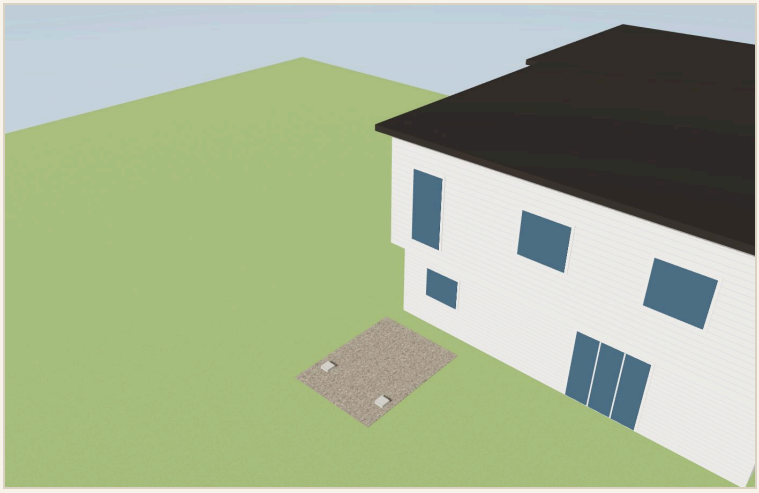
5 JOIST TAPE + BLOCKING



Nichigo G-Tape 3035 (black) on every top edge before decking, lapped over the ends and down the sides 1/2"; one row of 2x10 blocking over the beam, (3) structural screws each

6 FASTENING SCHEDULE

Ledger to house	LedgerLOK 3-5/8", 2 rows staggered 12" OC — engineer
Lateral	(4) DTT1Z, 1/2" rod into the floor framing
Joist to ledger	LUS28Z, 10d HDG every hole
Rim to ledger	HUCQ10-2-SDS, SDS screws
Rim plies	GRK RSS 3-1/8", 2 rows @ 12" staggered
Joist to beam	H2.5AZ each joist + rim, 10d x 1-1/2"
Beam plies	1/2" x 8" HDG bolts, 2 rows staggered @ 24"
Beam to post	CCQ76SDS2.5 cap, SDS screws
Post to pier	ABA66Z on the Diamond Pier, SD screws
Knee braces	1/2" through-bolt each end (x10 / x12 / x6)
Field boards	CONCEALoc, every joist, every gap
Borders / fascia	Cap-Tor xd color-matched, pre-drilled
Rail posts	(4) HeadLOK 6" into blocking



Step 1 · Footings — 2 Diamond Pier DP-50/50 driven-pin footings

T-400 · Step 1 · Footings

GOES IN

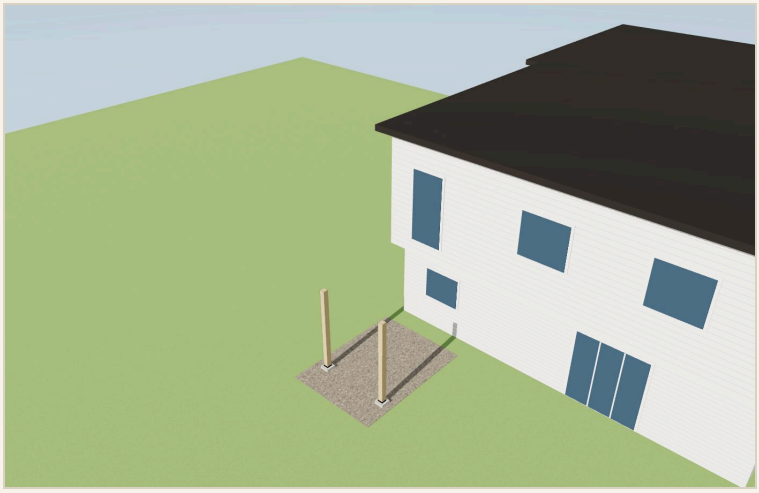
- (2) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: DROP (2)4x10 DF#1 at 9'-2" from the reference wall, posts at 9" / 7'-3".
- Set 2 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 Douglas fir #1 on ZMAX bases

T-401 · Step 2 · Posts

GOES IN

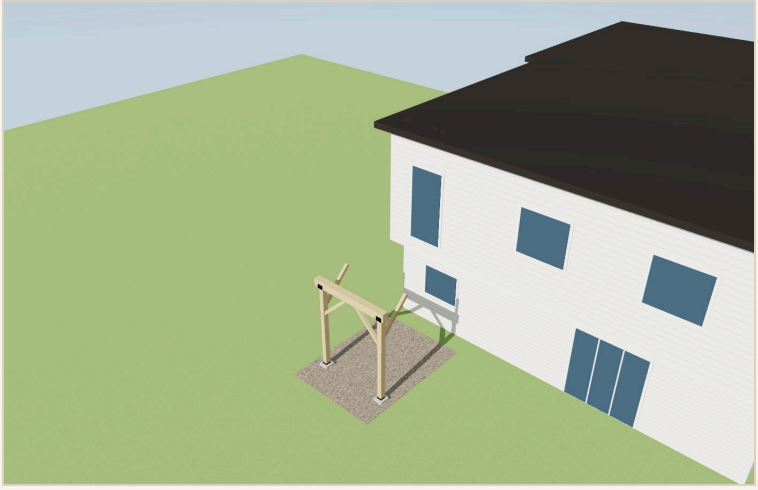
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt
- (1) Sail hardware: 4 stainless turnbuckles (jaw-jaw), D-shackles, snap hooks, 2 house anchor plates with through-bolts / structural screws into the framing

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): DROP (2)4x10 DF#1 posts 8'-0 1/16" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff



Step 4 · Beams — DROP (2)4x10 DF#1 (2 posts)

T-403 · Step 4 · Beams

GOES IN

- (2) 4x10x8 Douglas Fir #1
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal
- (2) 4x4x8 Douglas Fir #1 — knee braces
- (4) 1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)
- (2) 1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)
- (2) 1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)
- (10) 1/2" x 8" HDG carriage bolt + nut + washer

HOW

- DROP (2)4x10 DF#1: 7'-8" one piece, 2 posts @ 6'-6" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.
- Two-ply beam: through-bolt the plies with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a pair 6" from each end — before it goes up.
- Knee braces once the beam is capped: 4x4 at 45°, 2'-6" legs, one along the beam toward mid-span and one toward the house up to a 2x10 block between two joists (set the block with the frame) — 1/2" through-bolt each end, then pull the temporary bracing.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened
- Every knee brace bolted both ends; posts still plumb



Step 5 · Frame — 2x10 joists at 16", rims, ledger, blocking over the beam

T-404 · Step 5 · Frame

GOES IN

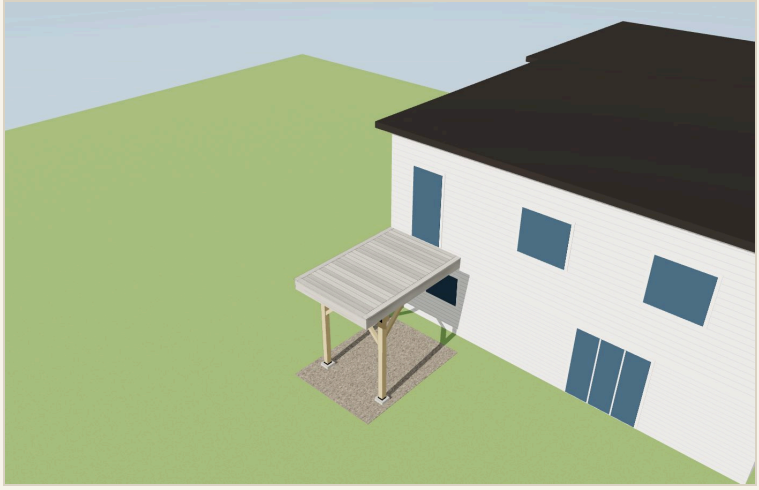
- (15) 2x10x12 #1 True Frame Joist SYP GC
- (14) Simpson LUS28Z face-mount hanger 2x8/2x10
- (4) Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)
- (1) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (1) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (9) Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)
- (1) Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams
- (2) Nichigo G-Tape 3035 2" x 65' black

HOW

- 2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
- A: 5 joists @ 16" OC x 11'-1". Crown up, hangers at the ledger.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 6 · Decking — TimberTech Reserve Driftwood, hidden fasteners

T-405 · Step 6 · Decking

GOES IN

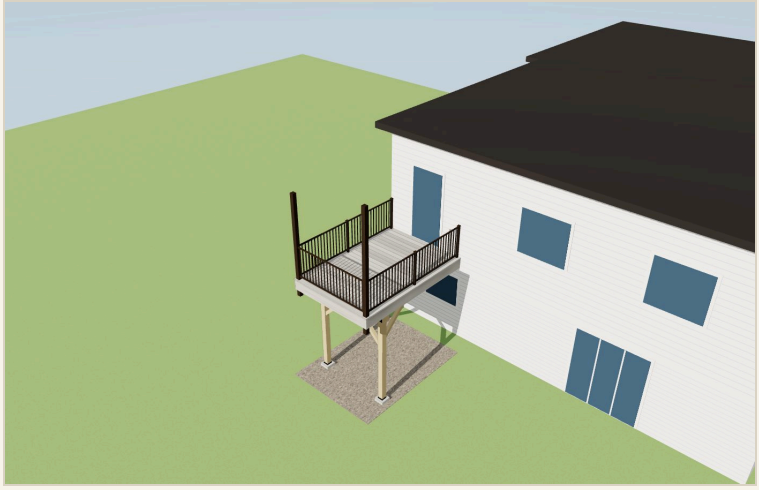
- (12) TimberTech Reserve Driftwood 1x6x16 Grooved
- (1) TimberTech Reserve Driftwood 1x6x12 Square Edge — borders
- (1) TimberTech Reserve Driftwood 1x6x20 Square Edge — borders
- (3) TimberTech Reserve Fascia Driftwood 0.58x11.95x12
- (1) TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)
- (1) Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct

HOW

- CONCEALoc at every joist in every gap (168)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
- Rows per zone: A 24. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · Rail — IRX Baluster 36" Bronze

T-406 · Step 7 · Rail

GOES IN

- (4) TimberTech Impression Rail Express 6' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze
- (1) TimberTech Impression Rail Express 8' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze
- (4) TimberTech Impression Rail Express 2" aluminum post kit 36" (post, base, cap, skirt), Bronze
- (1) TimberTech Bronze touch-up paint (cut ends)

HOW

- IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction

3D MODEL



The interactive 3D model lives at the online version of this document.

Prices: rate card · GC at cost · 37.0% GM (owner-set, under the 40% floor) · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.

GS EXTERIOR EXPERTS · DECK DIVISION · PERMIT SET

5549 S Lansing Ct · Elevated Deck

5549 S Lansing Ct, Englewood · Rachel Chiu · 8' x 12' deck, 10' above the yard, off the existing door · TrueFrame 2x10 @ 16" on a (2)4x10 Douglas fir beam and (2) 6x6 DF #1 posts on Diamond Piers · TimberTech Reserve Driftwood, single picture frame, CONCEALoc · Impression Rail Express, vertical balusters, modern top rail, Bronze · square shade sail on two steel posts at the rail corners · October 19, 2026

SHEET INDEX

1. Elevated deck · **G-001** General notes · design criteria
2. Elevated deck · **C-100** Site plan
3. Elevated deck · **A-002** Isometric — assembled, labeled
4. Elevated deck · **A-003** Isometric — exploded, labeled
5. Elevated deck · **S-100** Foundation plan
6. Elevated deck · **S-101** Framing plan
7. Elevated deck · **S-102** Structure isometric, labeled
8. Elevated deck · **A-101** Decking plan
9. Elevated deck · **A-201** Railing plan + elevation
10. Elevated deck · **A-202** Elevations
11. Elevated deck · **A-301** Section + front elevation
12. Elevated deck · **D-501** Details
13. Elevated deck · **D-502** Connection details — frame
14. Elevated deck · **D-504** Connection details — deck edge, rail, lateral

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. This set carries no pricing.

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 · PERMIT SET

PROJECT

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech Reserve Driftwood, IRX bronze)
5549 S Lansing Ct, Englewood CO 80111
Prepared for Rachel Chiu
92 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; 30 psf ground snow, roof 30 psf non-reducible (Arapahoe County design criteria); deck tables at 40 psf live + 10 psf dead
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult (3-second gust), Exposure B (suburban), Risk Cat. II — county design-criteria table (2015 cycle; confirm the 2021 table)
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC / IBC as adopted by Arapahoe County (Res. 21-394, eff. 4/1/2022); IRC R507 decks with the county-amended Colorado ICC Deck Building Guide; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces
Beams	DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 front and sides
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Decking	TimberTech Reserve Bronze — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

SHEET INDEX

G-001	General notes · design criteria
C-100	Site plan
A-001	Isometric · renders
A-002	Isometric — assembled, labeled
A-003	Isometric — exploded, labeled
S-100	Foundation plan
S-101	Framing plan
S-102	Structure isometric, labeled
A-101	Decking plan
A-201	Railing plan + elevation
A-202	Elevations
A-301	Section + front elevation
D-501	Details

SITE

Rear / front lot line: deck edge 13.0' from the line, setback per the PUD...
Side lot line: deck edge 23.7' from the line, 5' required
Easement: Check plat R1535678 for rear / side utility and drainage...
Unincorporated Arapahoe County (Cherry Creek CDP; Englewood mailing address). Parcel 2075-14-4-05-001 / PIN 031772427, Lot 41 Blk 9 Cherry Creek Vista 6th Flg Amend Resub (plat R1535678), 6,401 SF corner lot (front S Lansing Ct east, street side E Berry Ave north). R-P PUD Cherry Creek Vista (Z75-030 / P75-027 / P75-028). Deck on the WEST (rear) wall, north end, hung on the 2' second-floor cantilever that runs the full 33' wall (assessor sketch). Tax: CO 2.9 + RTD 1.0 + CD 0.1 + Arapahoe 0.25 = 4.25% (no city).

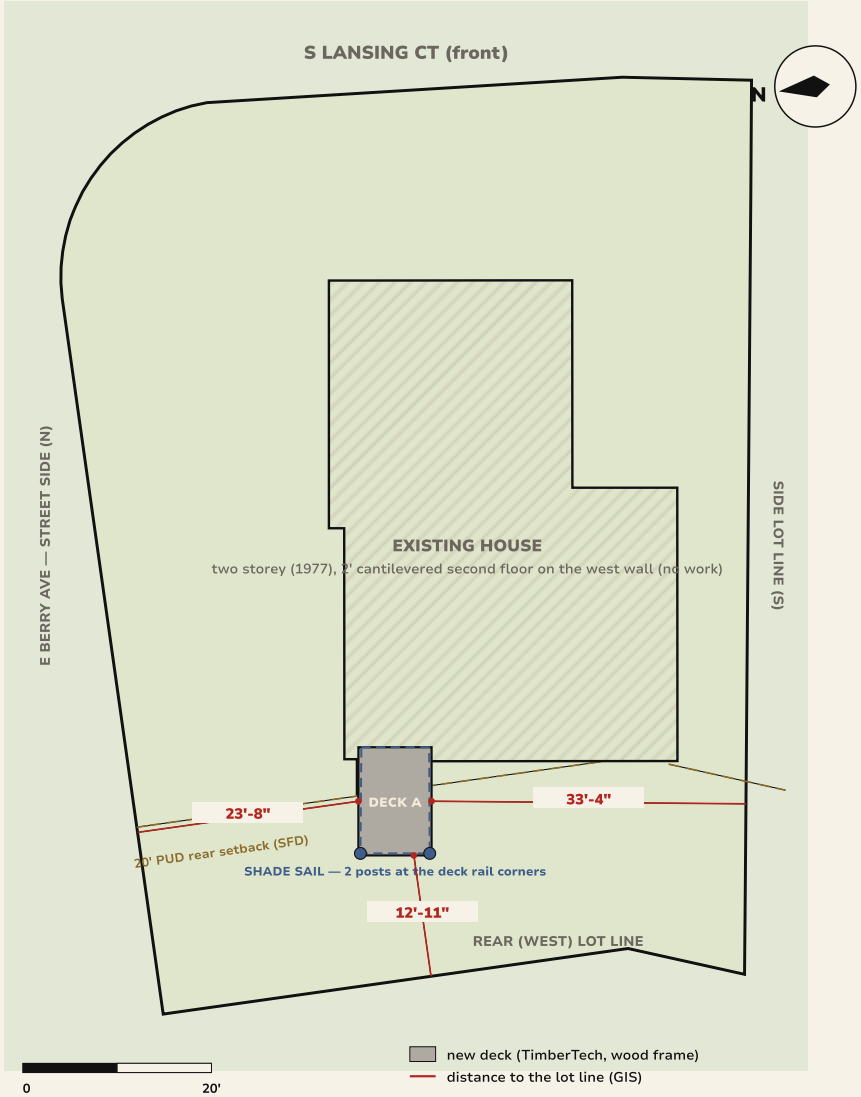
Elevated deck · C-100 · Site plan

C-100 **SITE PLAN**

Parcel, existing house, the new deck, distances to the lot lines, setbacks

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PARCEL

5549 S Lansing Ct, Englewood CO 80111
Parcel 2075-14-4-05-001 · PIN 031772427
Lot 41, Block 9, Cherry Creek Vista 6th Filing Amended Resub (plat R1535678)
Unincorporated Arapahoe County · zoning R-P PUD (Z75-030 / P75-027 / P75-028)
Lot 6,401 SF (GIS) · corner lot · flood zone X
House 1977, 2 storey, 1,704 SF + 660 SF basement + 441 SF garage (assessor)
HOA: Cherry Creek Vista II

LOT COVERAGE

Lot area	6,401 SF
Lot coverage limit	none in the PUD plan
Existing deck (assessor)	48 SF (8' x 6') — removed
New deck	8' x 12' · ~92 SF of frame, 10' above grade

SETBACKS

Elevated decks get no setback projection allowance (Arapahoe LDC) — the deck must meet the setbacks
GIS puts the new deck front ~13' off the west line: passes the 10' standard, not the 20' — confirm the governing column and the rear line (corner lot) with Zoning before submittal
E Berry Ave street side 10' required — ~23'-8" clear; south side 5' — ~33' clear
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the existing 8' x 6' deck and its two posts / pads; haul off
· Two Diamond Pier DP-50 foundations (no excavation spoils), 6x6 DF #1 posts with 4x4 knee braces
· New ledger on the 2' cantilevered second floor per the engineer's detail; membrane + Z-flash, (4) DTT1Z lateral ties
· No stairs — the deck is reached from the house door only
· Call 811 before the piers; protect the lawn and the patio slab at the north wall

Arapahoe County GIS (address point, parcel, 2024 roof prints, R-P zoning, jurisdiction, flood, wildfire layers), county assessor record and sketch, 2nd Amended PDP P75-027 setback table, county deck permit guide (rev. 8/20/2024) and the county-amended ICC deck guide. GIS lines are not a survey — an ILC governs if Zoning asks.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

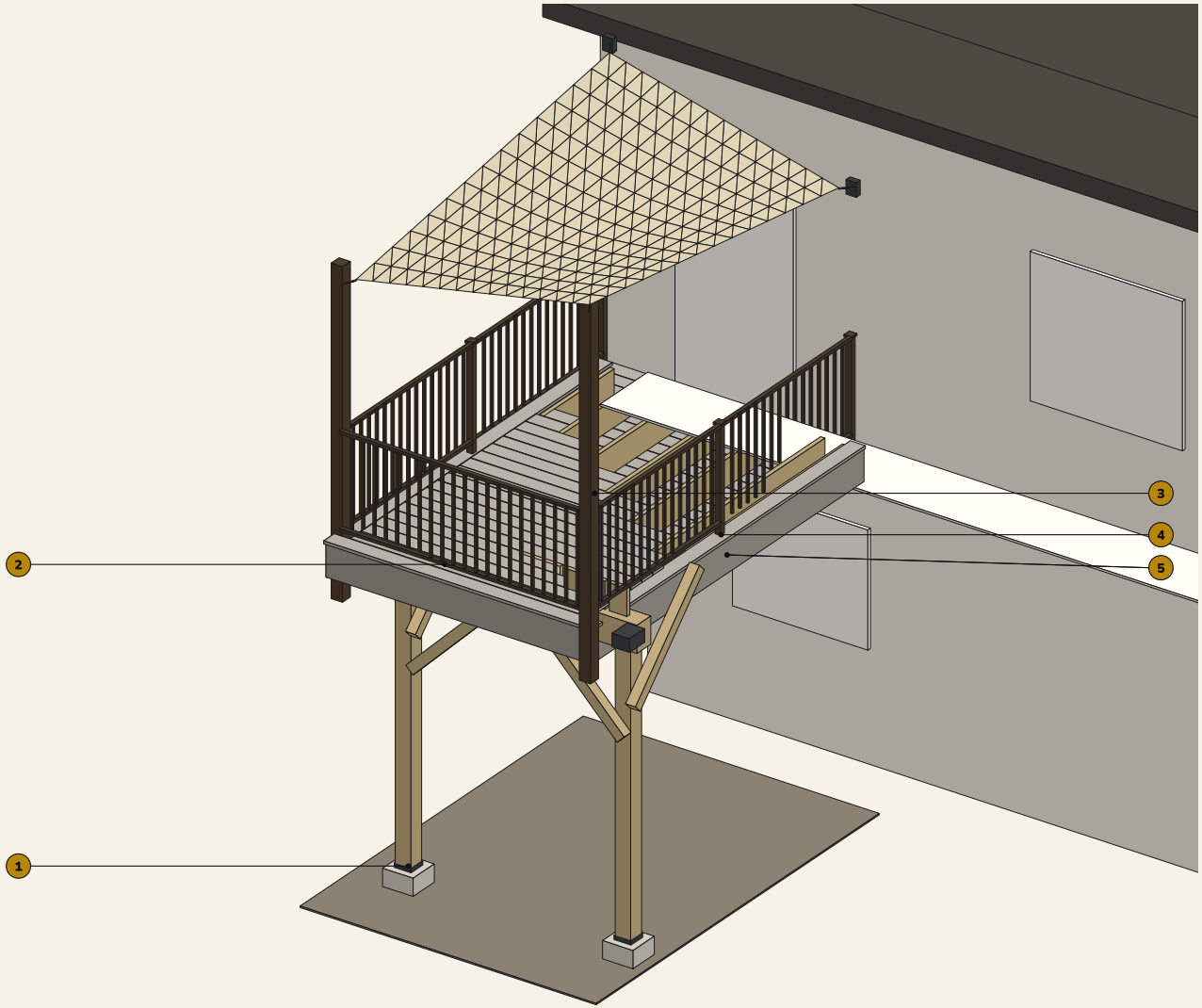
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
3. TimberTech Reserve Driftwood field decking, hidden fasteners
4. Picture frame + breaker boards — TimberTech Reserve Driftwood
5. Fascia

House shown for context only (the deck work only). The frame under the decking is called out on S-102 and A-003.

Elevated deck · A-003 · Isometric — exploded, labeled

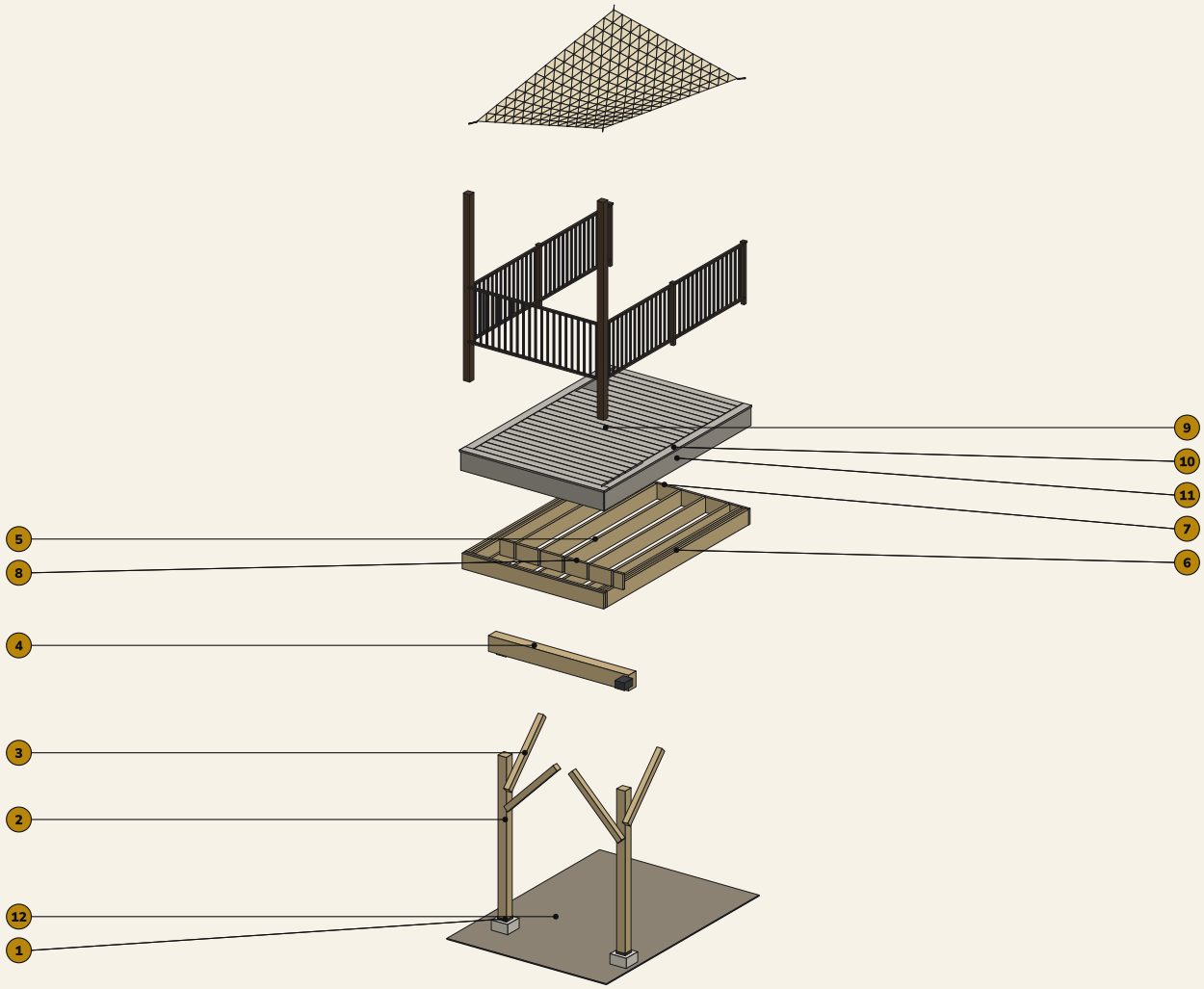
A-003

ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts + knee braces · beams · frame · decking · rail

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. TimberTech Reserve Driftwood field decking, hidden fasteners
10. Picture frame + breaker boards — TimberTech Reserve Driftwood
11. Fascia
12. Gravel over 6-mil poly under the decks

Layers lifted in build order. Lifts are for clarity only; every member returns straight down to its place.

Axonometric · not to scale · the model the plans are drawn from

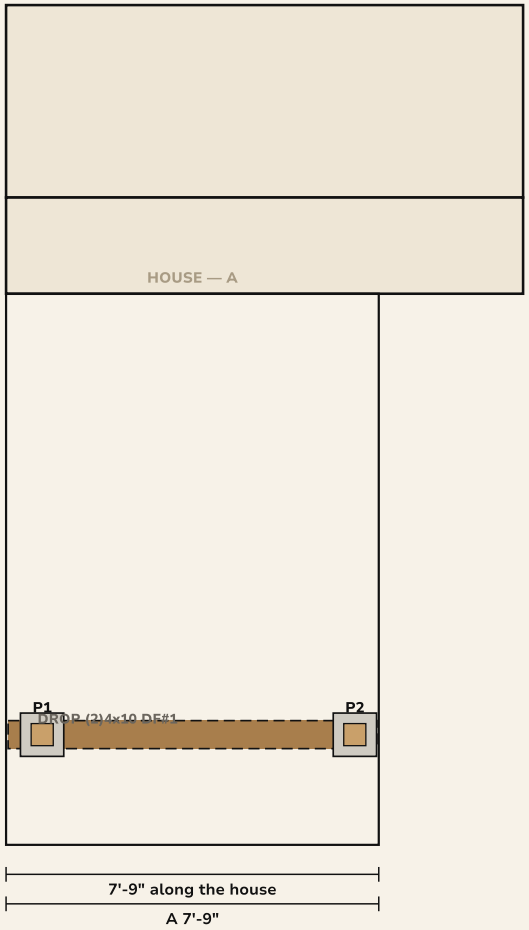
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	9"	9'-2"	A	2,214 lb
P2	7'-3"	9'-2"	A	2,214 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 2,214 lb

Plan · not to scale · house at the top · dimensions govern

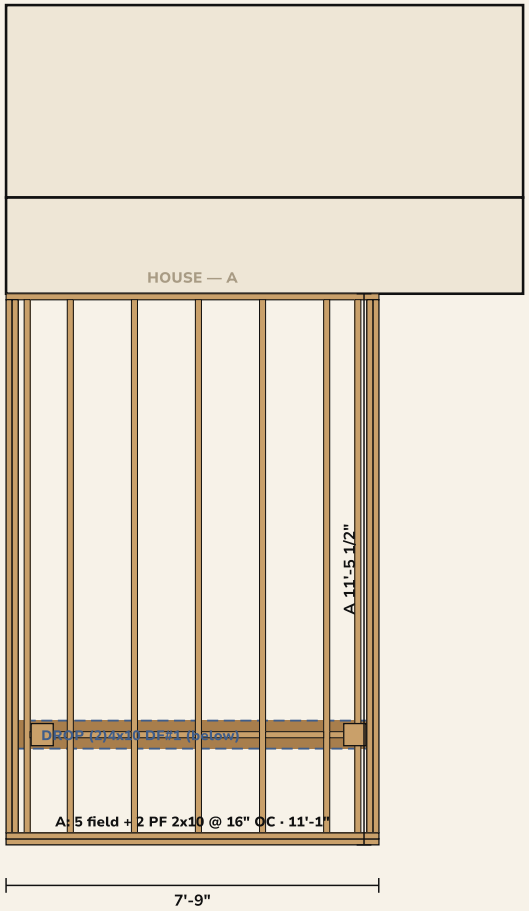
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRAMING NOTES

7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"

2-ply 2x10 front and sides

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

blocking one row over the beam, staggered

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Connectors:

1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post): 4

1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam): 2

1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block): 2

Simpson LUS28Z face-mount hanger 2x8/2x10: 14

Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 4

Simpson DTT1Z lateral tie kit w/ SDS screws: 4

Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden): 9

Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal:

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · S-102 · Structure isometric, labeled

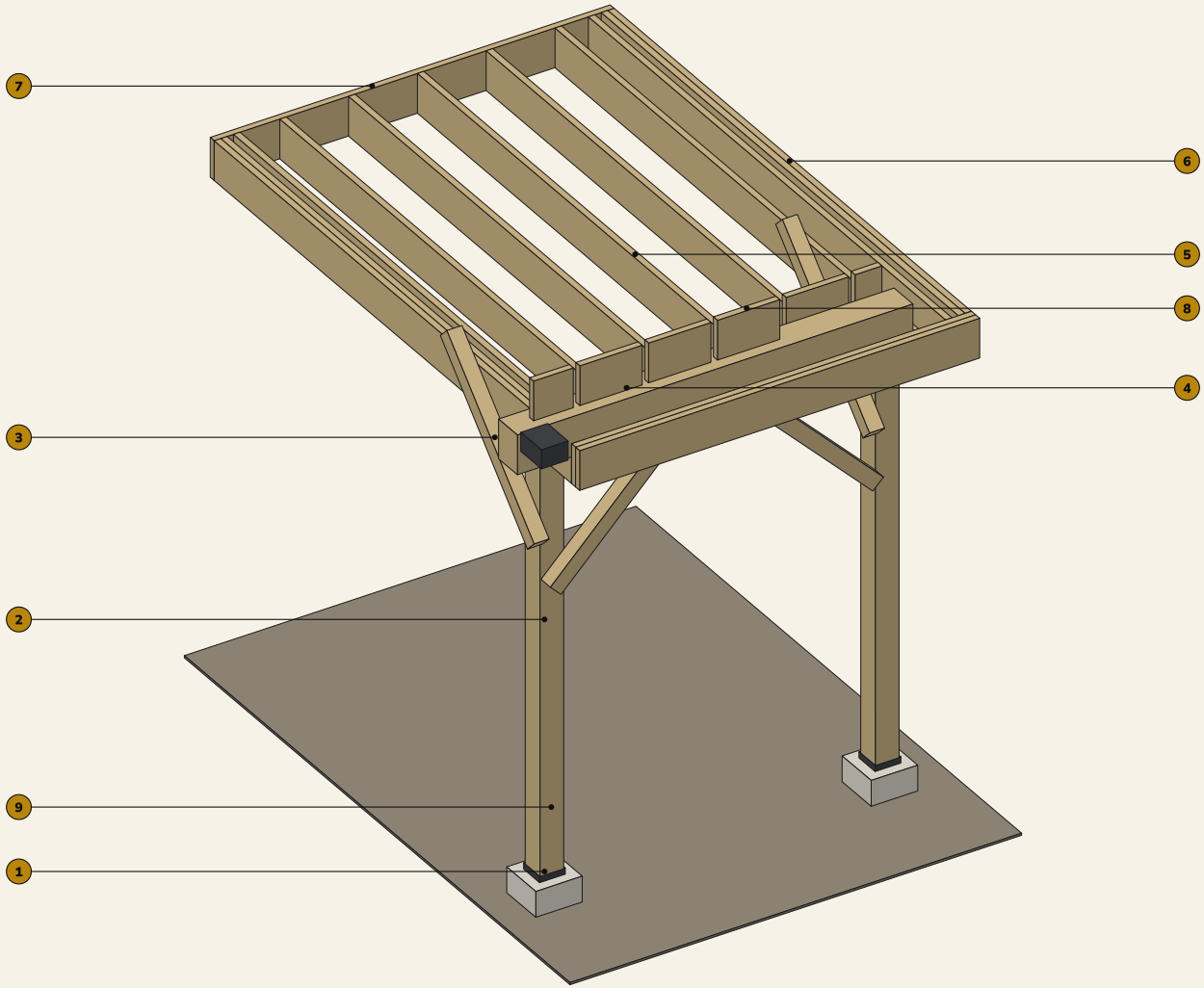
S-102

STRUCTURE ISOMETRIC — FRAME, LABELED

Piers, post bases, posts, knee braces, post caps, beam, joists, double rims, ledger, blocking

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. Gravel over 6-mil poly under the decks

Decking, rail and house removed to show the frame. Joists hang from the ledger in face-mount hangers, bear on the drop beam with H2.5AZ ties; double rims in concealed-flange hangers.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-101 · Decking plan

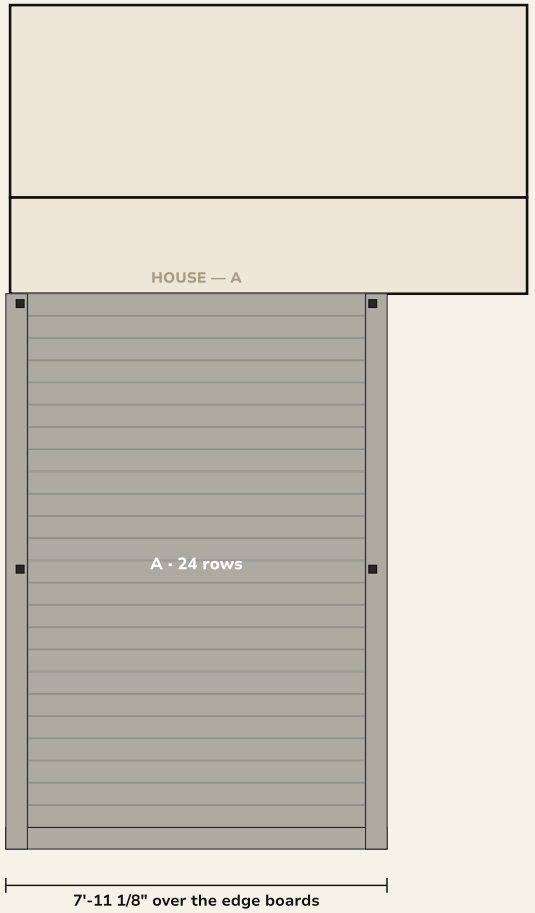
A-101

DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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DECKING

Field: TimberTech Reserve Driftwood, 5.36" x 0.94", 3/16" gaps

Border: Reserve Driftwood

Boards run parallel to the house; edge overhang 1 1/16" over the fascia

CONCEALoc at every joist in every gap (168)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per zone: A 24 (no rips)

Plan · not to scale · house at the top · dimensions govern

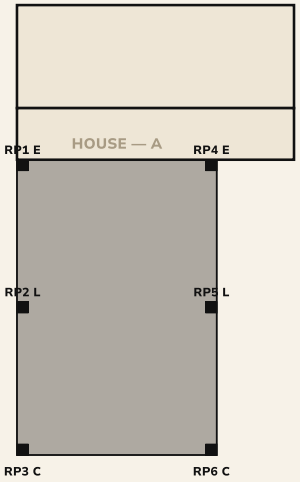
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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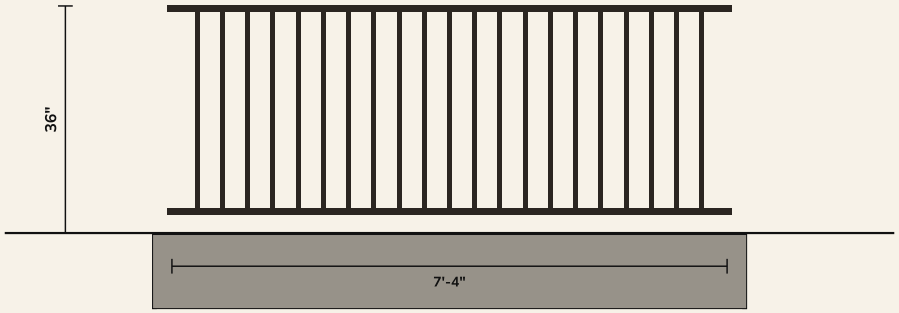
RAILING

IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

Bays: left 5'-6 1/4" · left 5'-6 1/4" · right 5'-6 1/4" · right 5'-6 1/4" · front 7'-4"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

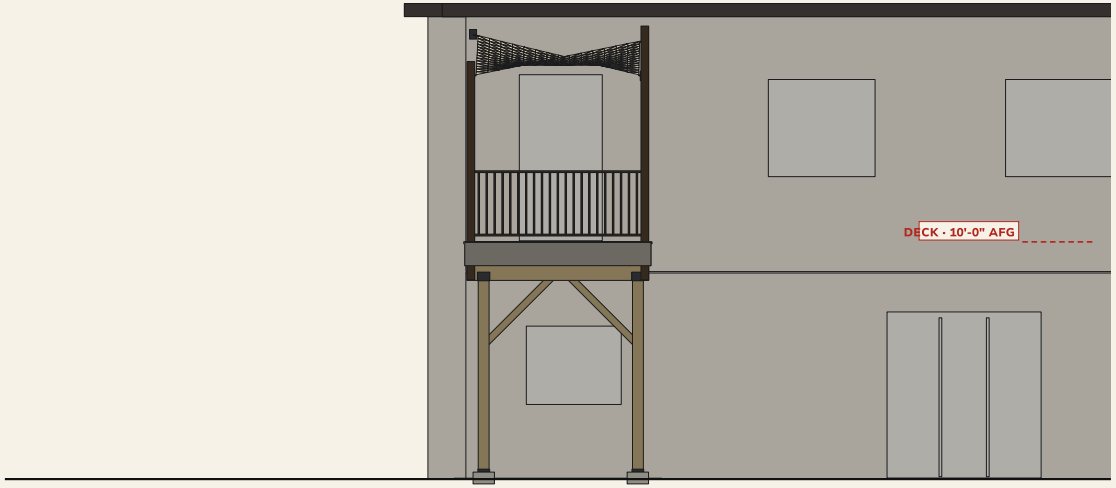
A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

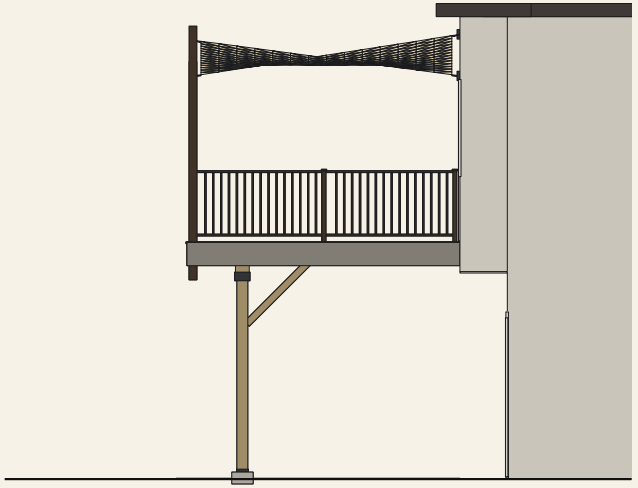
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRONT ELEVATION — from the yard



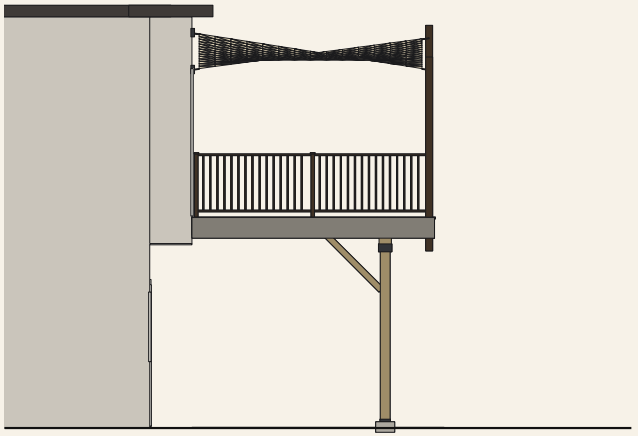
RIGHT END ELEVATION



NOTES

· Deck surface 10'-0" above grade
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Deck frame: 2x10 #1 True Frame GC @ 16" OC on a (2)4x10 DF #1 drop beam, 2 posts with 4x4 knee braces both ways
· Guard 36" min. on every open side — the surface is 10'-0" above grade (IRC R312.1)

LEFT END ELEVATION



Elevated deck · A-301 · Section + front elevation

A-301

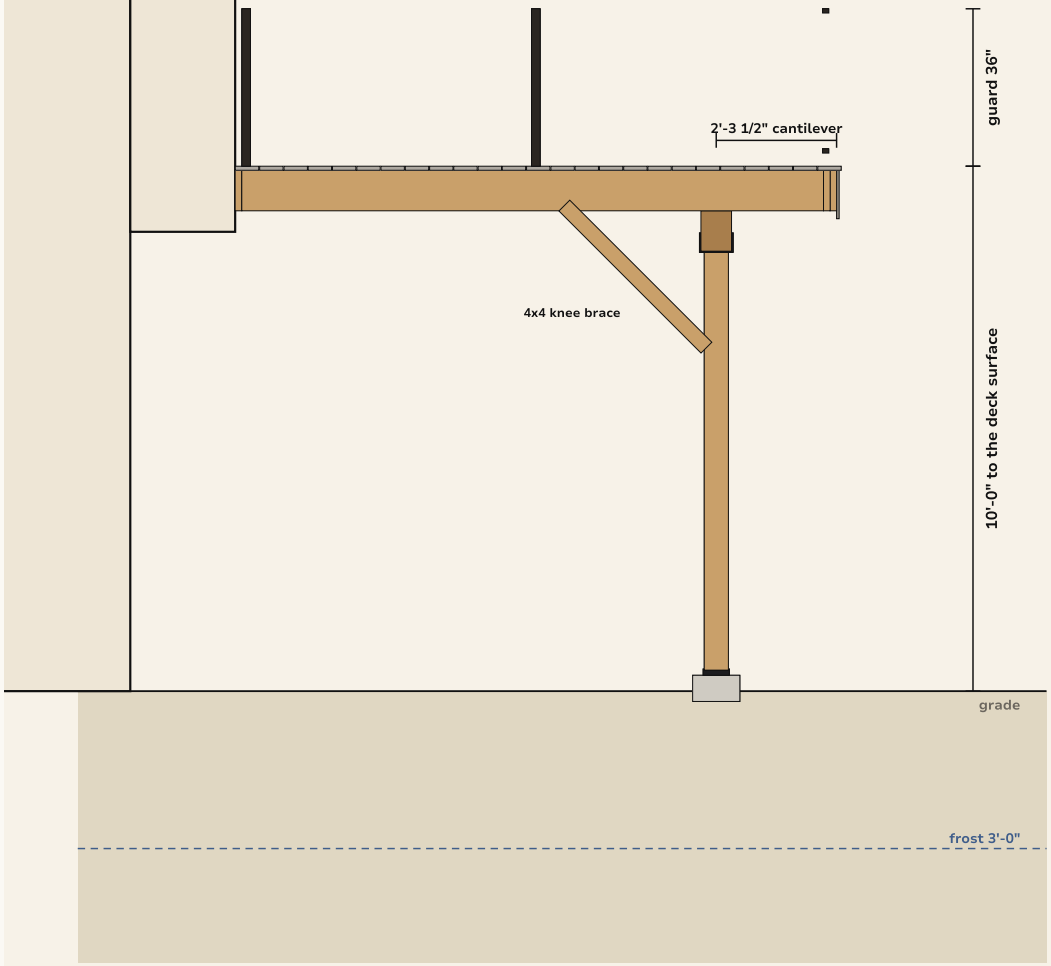
SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

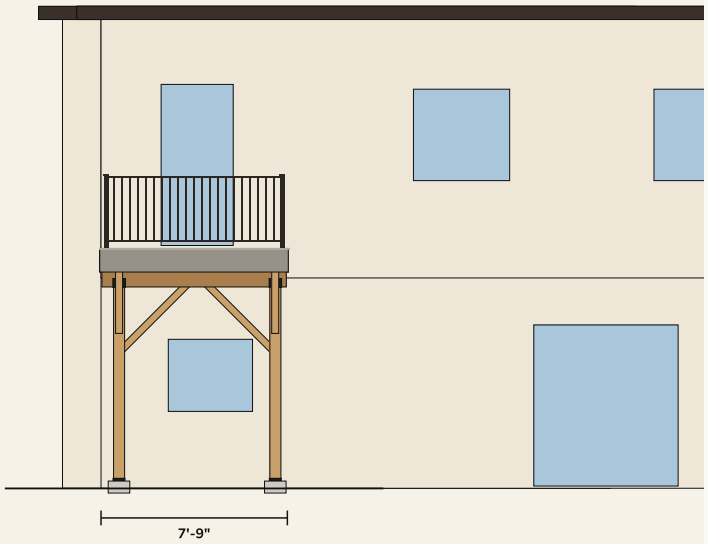
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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SECTION through zone A at 3'-10 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Posts 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · D-501 · Details

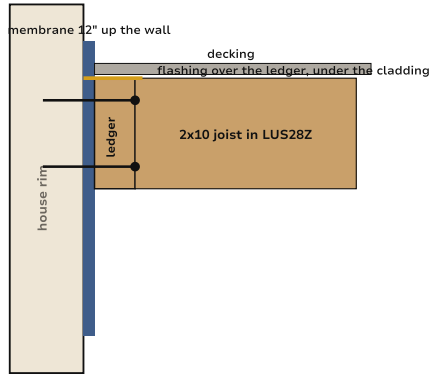
D-501 **DETAILS**

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

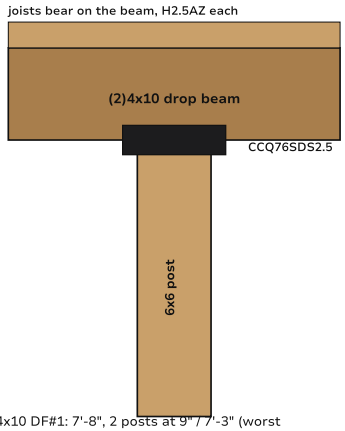
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1 LEDGER AT THE HOUSE



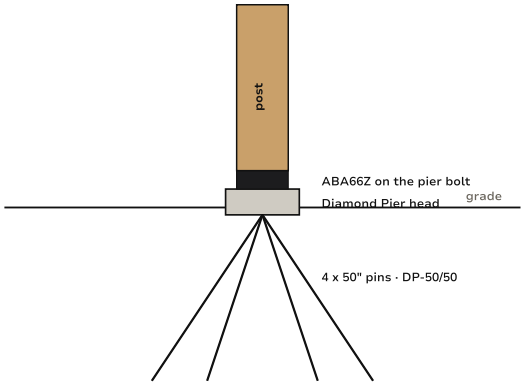
2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

2 POST · BEAM · CAP

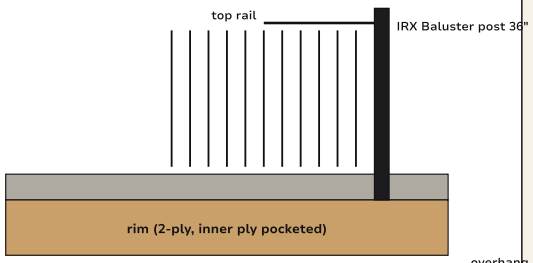


DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

3 FOOTING · POST BASE

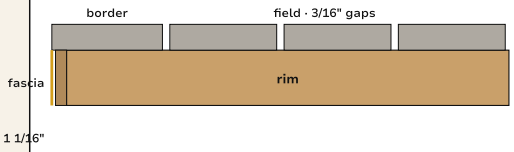


4 RAIL POST · GUARD



IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

5 DECK EDGE · BORDER



CONCEALoc at every joist in every gap (168)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.

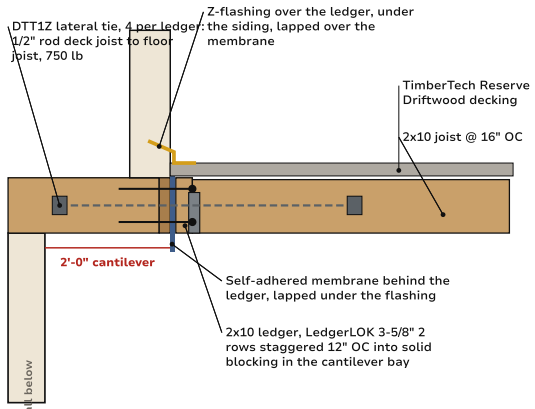
D-502 CONNECTION DETAILS — FRAME

Ledger on the cantilever · hangers · joists on the beam · (2)4x10 on the cap · pier + base · knee braces

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

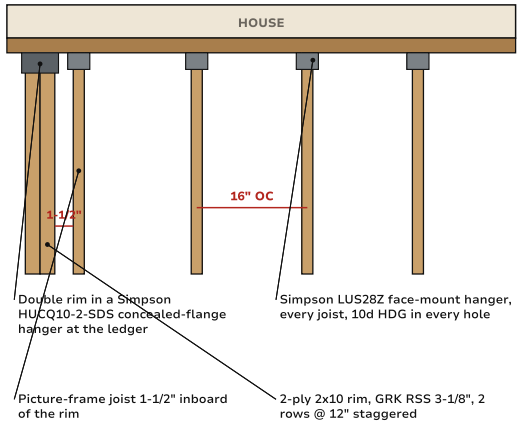
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1 LEDGER ON THE CANTILEVERED FLOOR — SECTION

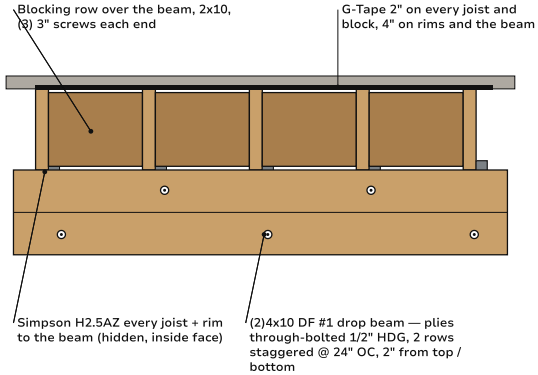


ENGINEER: Arapahoe County bars a ledger on a cantilevered floor without engineer approval. The stamped detail governs: ledger into the blocked cantilever joists, or a freestanding beam on posts at the house.

2 HANGERS — PLAN AT THE LEDGER + SIDE RIM



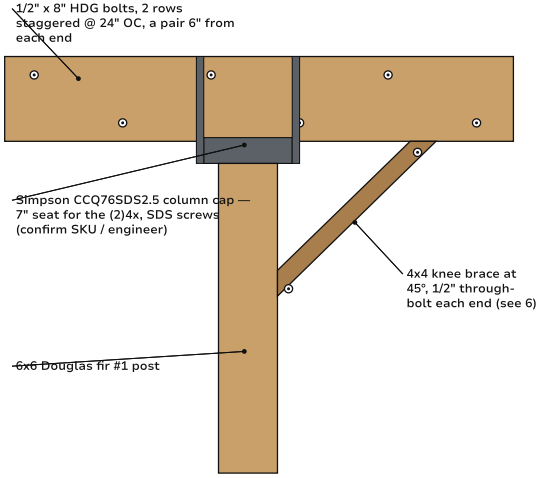
3 JOISTS ON THE DROP BEAM — SECTION



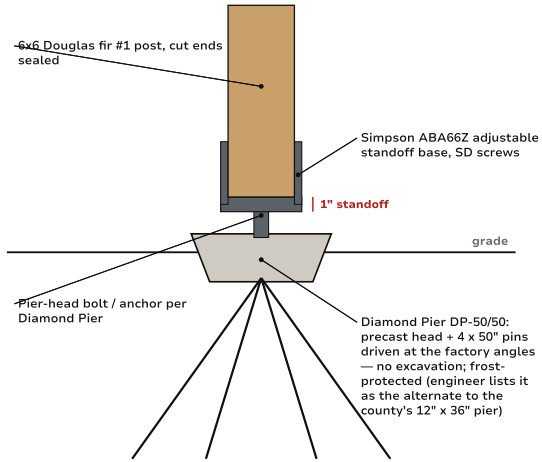
Joists bear the full beam width

Cantilever 2'-3 1/2" past the beam — over L/4: engineer

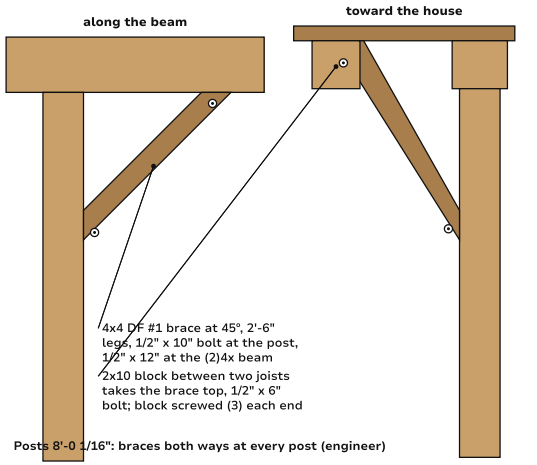
4 (2)4x10 BEAM ON THE 6x6 — ELEVATION



5 POST BASE ON THE DIAMOND PIER — SECTION



6 KNEE BRACES — BOTH DIRECTIONS



Posts 8'-0 1/16": braces both ways at every post (engineer)

Elevated deck · D-504 · Connection details — deck edge, rail, lateral

D-504

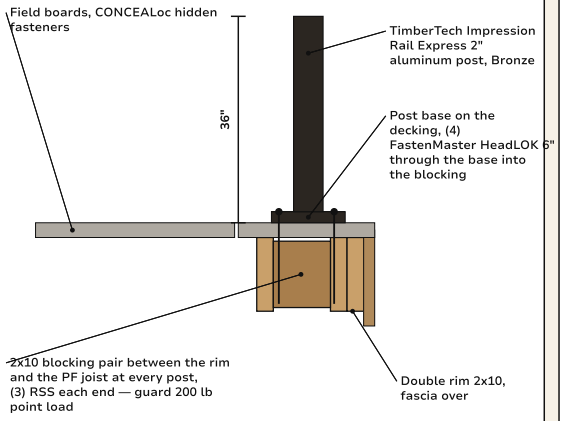
CONNECTION DETAILS — DECK EDGE, RAIL, LATERAL

Rail post mounting · picture frame + fascia · guard bay · DTT1Z lateral ties · joist tape · fastening schedule

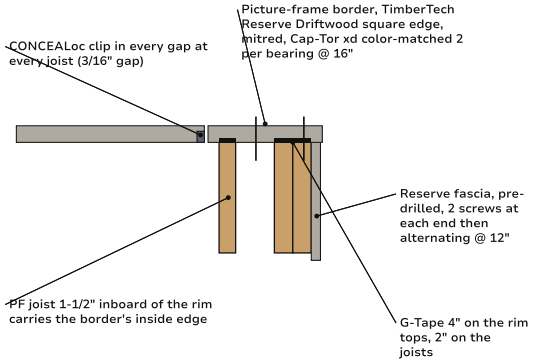
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 · PERMIT SET

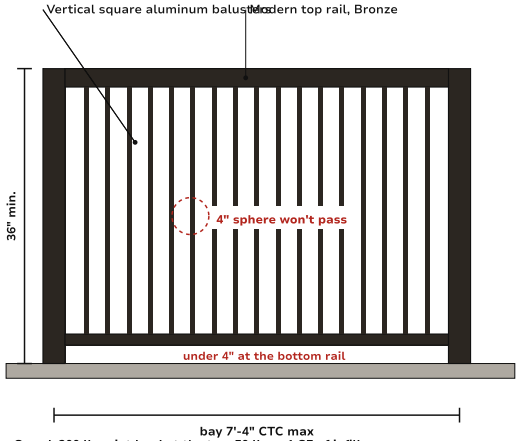
1 RAIL POST — SURFACE MOUNT ON BLOCKING



2 DECK EDGE — PICTURE FRAME + FASCIA

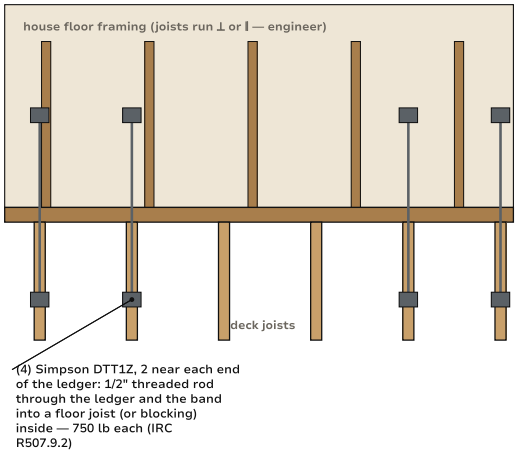


3 GUARD — ONE BAY, ELEVATION

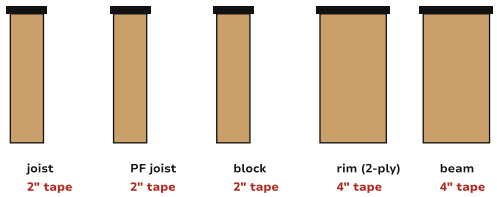


Guard: 200 lb point load at the top, 50 lb on 1 SF of infill (IRC Table R301.5) — submit the rail code report

4 LATERAL LOAD — DTT1Z TIES (PLAN)



5 JOIST TAPE + BLOCKING

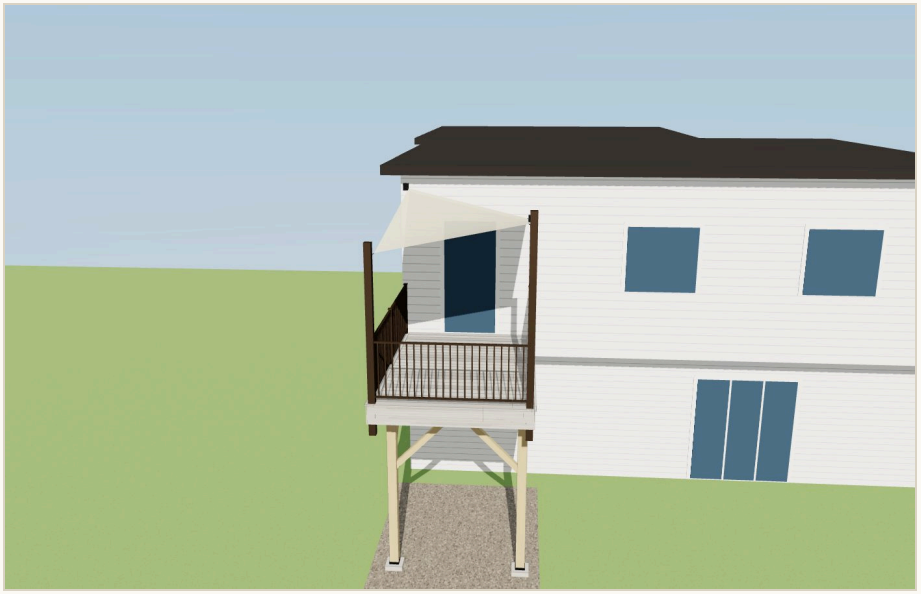


Nichigo G-Tape 3035 (black) on every top edge before decking, lapped over the ends and down the sides 1/2"; one row of 2x10 blocking over the beam, (3) structural screws each

6 FASTENING SCHEDULE

Ledger to house	LedgerLOK 3-5/8", 2 rows staggered 12" OC — engineer
Lateral	(4) DTT1Z, 1/2" rod into the floor framing
Joist to ledger	LUS28Z, 10d HDG every hole
Rim to ledger	HUCQ10-2-SDS, SDS screws
Rim plies	GRK RSS 3-1/8", 2 rows @ 12" staggered
Joist to beam	H2.5AZ each joist + rim, 10d x 1-1/2"
Beam plies	1/2" x 8" HDG bolts, 2 rows staggered @ 24"
Beam to post	CCQ76SDS2.5 cap, SDS screws
Post to pier	ABA66Z on the Diamond Pier, SD screws
Knee braces	1/2" through-bolt each end (x10 / x12 / x6)
Field boards	CONCEALoc, every joist, every gap
Borders / fascia	Cap-Tor xd color-matched, pre-drilled
Rail posts	(4) HeadLOK 6" into blocking

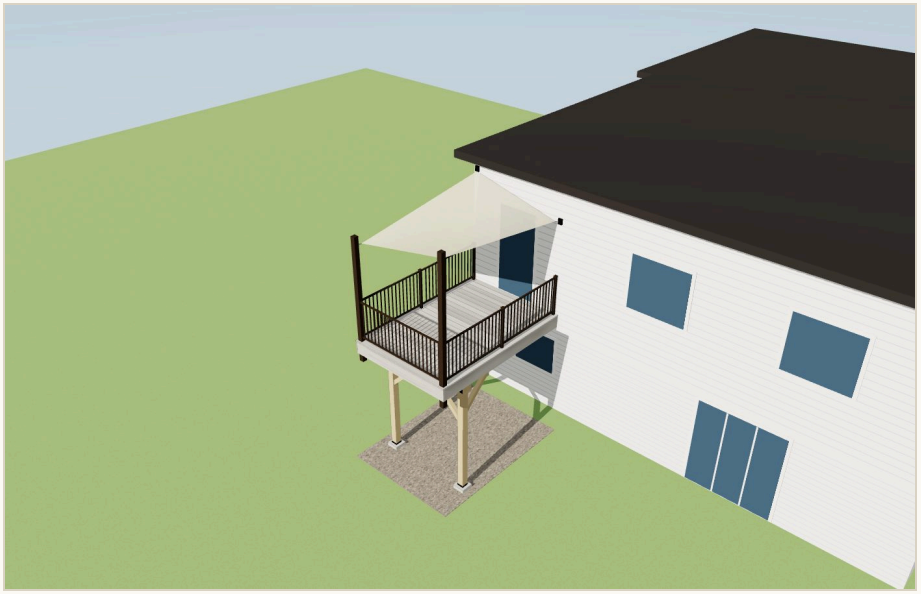
Deck



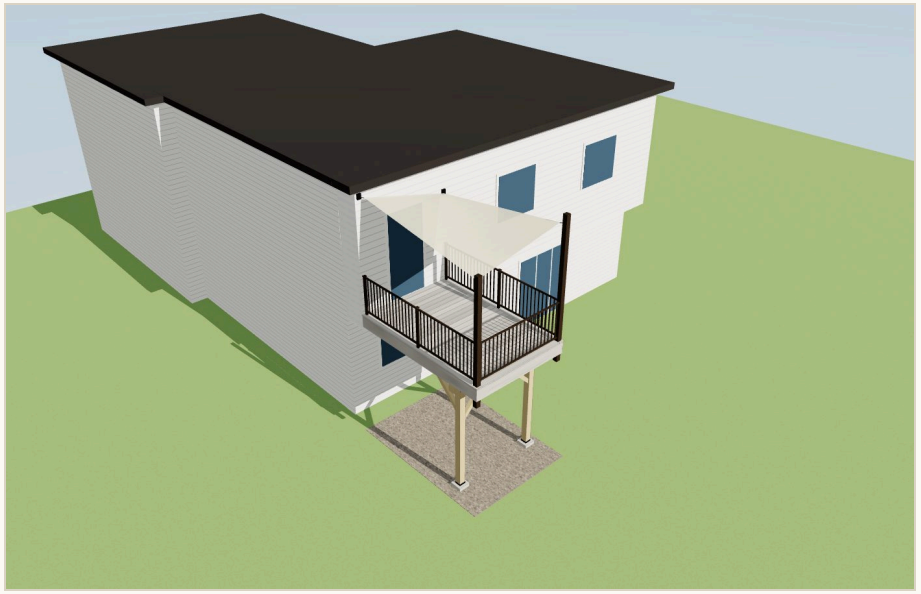
From the yard



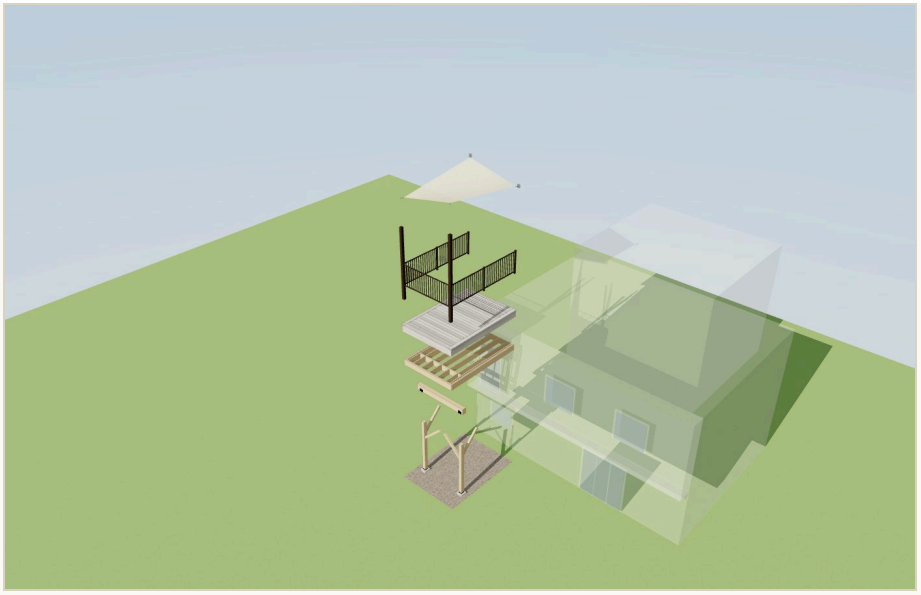
From the front corner



Isometric



Isometric — opposite corner



Exploded — every layer



Exploded — from the corner



Structure from below



Under the deck

GS EXTERIOR EXPERTS · DECK DIVISION · MATERIAL TAKE-OFF + BUILD SET · FOR DECKS & DOCKS

5549 S Lansing Ct · Elevated Deck

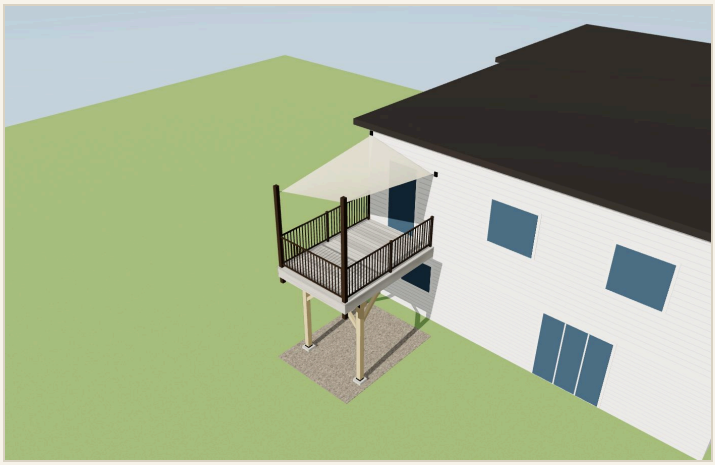
5549 S Lansing Ct, Englewood · Rachel Chiu · 8' x 12' deck, 10' above the yard, off the existing door · TrueFrame 2x10 @ 16" on a (2)4x10 Douglas fir beam and (2) 6x6 DF #1 posts on Diamond Piers · TimberTech Reserve Driftwood, single picture frame, CONCEALoc · Impression Rail Express, vertical balusters, modern top rail, Bronze · square shade sail on two steel posts at the rail corners · October 19, 2026

Account GSEXT1 · ship to 5549 S Lansing Ct, Englewood, CO 80111 · please quote every line; call Jade Pender 303-550-9558 with substitutions.

SQUARE FEET	FOOTINGS	ORDER LINES	HEIGHT
92	**2**	**40**	**10'-0"**

THE JOB

Deck	92 SF · 10'-0" above grade · 7'-9" x 11'-5 1/2" outside to outside
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 front and sides
Beams	DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)
Footings / posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50
Decking	TimberTech Reserve Bronze — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF
Stairs	none



Isometric from the stair corner



Isometric from the open corner

MATERIAL TAKE-OFF NET = counted off the drawings · ORDER = what to pull and ship

Lumber

NET	ORDER	UNIT	ITEM	WHY
15	**16**	ea	2x10x12 #1 True Frame Joist SYP GC	+1 cull (5 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · 2 blocking, front rim ply · knee-brace block, ledger, rail post block · rail post block)
2	**2**	ea	4x10x8 Douglas Fir #1	exact (2 drop beam: DROP (2)4x10 DF#1)
2	**2**	ea	6x6x10 Douglas Fir #1	exact (yields 2 posts at ~8'-0 1/16")
2	**3**	ea	4x4x8 Douglas Fir #1 — knee braces	+1 (4 braces at 3'-9 15/16", 45° plumb cuts, 2'-6" legs: one along the beam, one up to the joists at every post)

Footings

NET	ORDER	UNIT	ITEM	WHY
2	**2**	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock
2	**2**	ea	Simpson ABA66Z adjustable post base 6x6 on the pier bolt	exact

Hardware

NET	ORDER	UNIT	ITEM	WHY
4	**5**	ea	1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)	+1
2	**3**	ea	1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)	+1
2	**3**	ea	1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)	+1
14	**14**	ea	Simpson LUS28Z face-mount hanger 2x8/2x10	exact (7 joists x 2 ends)
4	**4**	ea	Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)	exact (2 side rims x 2 ends) — confirm stock
1	**1**	box	Camo 9ga HDG hanger nail 3" 5 lb (~350)	84 needed of ~350
1	**1**	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	146 needed of ~650
1	**1**	box	FastenMaster LedgerLOK 3-5/8" 50 ct	9 needed
4	**4**	ea	Simpson DTT1Z lateral tie kit w/ SDS screws	exact (2 near each end of every ledger, into house floor framing)
9	**9**	ea	Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)	exact (7 joists + 2 rims at each drop beam)
2	**2**	ea	Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal	exact
10	**12**	ea	1/2" x 8" HDG carriage bolt + nut + washer	+2 ((2)4x beam plies through-bolted, 2 rows staggered @ 24" OC, 2" from top and bottom)
1	**1**	box	Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct	24 needed (bases 24 · caps 0)

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	GRK RSS #10 x 3-1/8" structural screw 100 ct	94 needed (laminations 70 · blocking 24)
1	**1**	box	FastenMaster HeadLOK 6" (50 ct) — rail posts 4 ea	24 needed (6 rail posts x 4)

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	**1**	set	Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams	exact
2	**2**	roll	Nichigo G-Tape 3035 2" x 65' black	93' needed of 130'
1	**1**	roll	Nichigo G-Tape 3035 4" x 65' black	38' needed of 65'

Decking

NET	ORDER	UNIT	ITEM	WHY
12	**14**	ea	TimberTech Reserve Driftwood 1x6x16 Grooved	+2 (A: 24 rows @ 7'-0", 2 per 16' board)
1	**1**	ea	TimberTech Reserve Driftwood 1x6x12 Square Edge — borders	exact (right border 11'-6 9/16")
1	**2**	ea	TimberTech Reserve Driftwood 1x6x20 Square Edge — borders	+1 (left border 11'-6 9/16" + front border 7'-11 1/8")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)	168 needed · 500 supplied
1	**1**	box	Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct	106 screws needed (first row 7 · border 58 · fascia 41)

Fascia

NET	ORDER	UNIT	ITEM	WHY
3	**4**	ea	TimberTech Reserve Fascia Driftwood 0.58x11.95x12	+1 (left fascia 11'-6 1/16" · right fascia 11'-6 1/16" · front fascia 7'-10 3/16")

Rail

NET	ORDER	UNIT	ITEM	WHY
4	**4**	ea	TimberTech Impression Rail Express 6' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze	exact (cut to 5'-3 3/4")
1	**1**	ea	TimberTech Impression Rail Express 8' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze	exact (cut to 7'-1 1/2")
4	**4**	kit	TimberTech Impression Rail Express 2" aluminum post kit 36" (post, base, cap, skirt), Bronze	exact (2 CORNER, 2 END, 2 LINE; 2 corners are the sail posts)
1	**1**	ea	TimberTech Bronze touch-up paint (cut ends)	exact

Shade sail

NET	ORDER	UNIT	ITEM	WHY
73	**73**	SF	Shade sail — custom commercial-grade HDPE shade sail, 4 corners, hypar twist, Sand, stainless D-rings, sewn-in perimeter cable	made to the measured attachment points, ~6% under for tensioning
1	**1**	ea	Steel sail post 4" x 4" x 1/4" HSS, galvanized + powder coat Bronze, pad eye welded at the top, rail-bracket tabs — 11' (9'-1 3/16" above the deck + 1'-7" down the rim)	post 1: the rail corner post at this corner
1	**1**	ea	Steel sail post 4" x 4" x 1/4" HSS, galvanized + powder coat Bronze, pad eye welded at the top, rail-bracket tabs — 10' (7'-7 3/16" above the deck + 1'-7" down the rim)	post 2: the rail corner post at this corner
6	**8**	ea	5/8" x 10" HDG through-bolt + nut + plate washers — sail posts through the double rim and blocking	+2 (3 per post)
1	**1**	ea	2x10x8 #1 True Frame — solid blocking behind each sail post (rim to PF joist and the next joist)	4 blocks
1	**1**	kit	Sail hardware: 4 stainless turnbuckles (jaw-jaw), D-shackles, snap hooks, 2 house anchor plates with through-bolts / structural screws into the framing	316 stainless, marine grade

FASTENER & CONNECTOR SCHEDULE

Knee braces	4x4 DF #1 at 45°, 2'-6" legs, both directions at every post: one along the beam toward mid-span, one toward the house up to a 2x10 block between the joists; (1) 1/2" through-bolt each end, 2" min. edge distance
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Beam	(2)4x plies through-bolted with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a bolt pair 6" from each end; CCQ cap with SDS screws at each post
Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	CONCEALoc at every joist in every gap (168)
Face screws	Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
Rail	IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

CUT LIST

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists	2x10	11'-1"	5	@ 16" OC, crown up
Picture-frame joists	2x10	11'-1"	2	centre 5 1/4" from each frame face

MEMBER	STOCK	LENGTH	QTY	NOTE
Side rim plies	2x10	11'-1"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies	2x10	7'-9"	2	inner ply takes the hangers; outer ply laminated after
Ledger	2x10	7'-9"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking	2x10	1'-2 1/2"	6	1 row at 9'-2" from the wall
Field boards run A	Reserve Driftwood	7'-0"	24	boards run parallel to the house
DROP (2)4x10 DF#1 beam	4x10	7'-8"	2	y 9'-2"; posts at 9" / 7'-3"
Posts under DROP (2)4x10 DF#1	6x6	8'-0 1/16"	2	field-measure each
Knee braces	4x4	3'-9 15/16"	4	45° both ends, 2'-6" legs · along the beam toward mid-span + up to a block between the joists
Left border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist
Right border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist
Front border	Reserve Driftwood square	7'-11 1/8"	1	mitre at the corners; dividers on the doubled joist
Left fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia	fascia	7'-10 3/16"	1	flush under the board nose; full board at the front corners, short piece at the house

G-001 General notes · design criteria

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PROJECT

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech Reserve Driftwood, IRX bronze)
5549 S Lansing Ct, Englewood CO 80111
Prepared for Rachel Chiu
92 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; 30 psf ground snow, roof 30 psf non-reducible (Arapahoe County design criteria); deck tables at 40 psf live + 10 psf dead
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult (3-second gust), Exposure B (suburban), Risk Cat. II — county design-criteria table (2015 cycle; confirm the 2021 table)
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC / IBC as adopted by Arapahoe County (Res. 21-394, eff. 4/1/2022); IRC R507 decks with the county-amended Colorado ICC Deck Building Guide; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces
Beams	DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 front and sides
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Decking	TimberTech Reserve Bronze — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

SHEET INDEX

G-001	General notes · design criteria
C-100	Site plan
A-001	Isometric · renders
A-002	Isometric — assembled, labeled
A-003	Isometric — exploded, labeled
S-100	Foundation plan
S-101	Framing plan
S-102	Structure isometric, labeled
A-101	Decking plan
A-201	Railing plan + elevation
A-202	Elevations
A-301	Section + front elevation
D-501	Details
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)

SITE

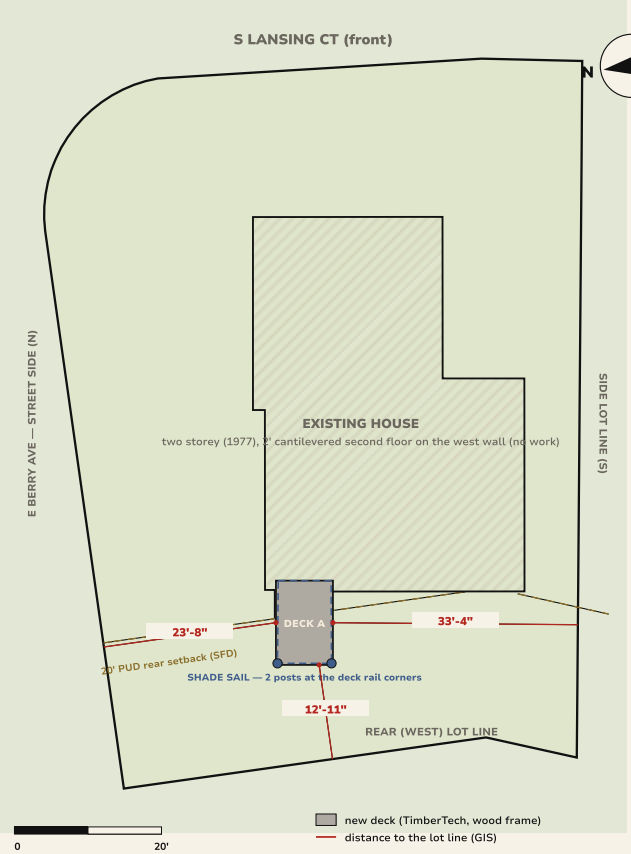
Rear / front lot line: deck edge 13.0' from the line, setback per the PUD...
Side lot line: deck edge 23.7' from the line, 5' required
Easement: Check plat R1535678 for rear / side utility and drainage...
Unincorporated Arapahoe County (Cherry Creek CDP; Englewood mailing address). Parcel 2075-14-4-05-001 / PIN 031772427, Lot 41 Blk 9 Cherry Creek Vista 6th Flg Amend Resub (plat R1535678), 6,401 SF corner lot (front S Lansing Ct east, street side E Berry Ave north). R-P PUD Cherry Creek Vista (Z75-030 / P75-027 / P75-028). Deck on the WEST (rear) wall, north end, hung on the 2' second-floor cantilever that runs the full 33' wall (assessor sketch). Tax: CO 2.9 + RTD 1.0 + CD 0.1 + Arapahoe 0.25 = 4.25% (no city).

C-100 SITE PLAN

Parcel, existing house, the new deck, distances to the lot lines, setbacks

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PARCEL

5549 S Lansing Ct, Englewood CO 80111
Parcel 2075-14-4-05-001 · PIN 031772427
Lot 41, Block 9, Cherry Creek Vista 6th Filing Amended Resub (plat R1535678)
Unincorporated Arapahoe County · zoning R-P PUD (Z75-030 / P75-027 / P75-028)
Lot 6,401 SF (GIS) · corner lot · flood zone X
House 1977, 2 storey, 1,704 SF + 660 SF basement + 441 SF garage (assessor)
HOA: Cherry Creek Vista II

LOT COVERAGE

Lot area	6,401 SF
Lot coverage limit	none in the PUD plan
Existing deck (assessor)	48 SF (8' x 6') — removed
New deck	8' x 12' · ~92 SF of frame, 10' above grade

SETBACKS

Elevated decks get no setback projection allowance (Arapahoe LDC) — the deck must meet the setbacks
GIS puts the new deck front ~13' off the west line: passes the 10' standard, not the 20' — confirm the governing column and the rear line (corner lot) with Zoning before submittal
E Berry Ave street side 10' required — ~23'-8" clear; south side 5' — ~33' clear
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the existing 8' x 6' deck and its two posts / pads; haul off
· Two Diamond Pier DP-50 foundations (no excavation spoils), 6x6 DF #1 posts with 4x4 knee braces
· New ledger on the 2' cantilevered second floor per the engineer's detail; membrane + Z-flash, (4) DTT1Z lateral ties
· No stairs — the deck is reached from the house door only
· Call 811 before the piers; protect the lawn and the patio slab at the north wall

Arapahoe County GIS (address point, parcel, 2024 roof prints, R-P zoning, jurisdiction, flood, wildfire layers), county assessor record and sketch, 2nd Amended PDP P75-027 setback table, county deck permit guide (rev. 8/20/2024) and the county-amended ICC deck guide. GIS lines are not a survey — an ILC governs if Zoning asks.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

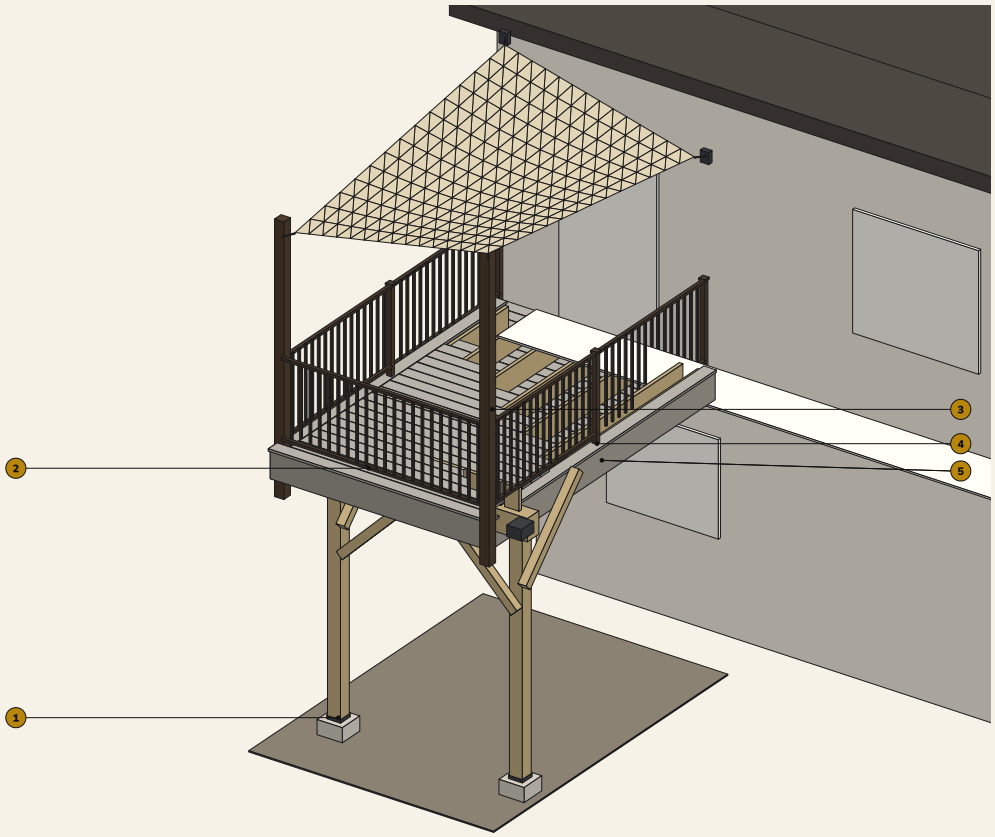
A-002 Isometric — assembled, labeled

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
3. TimberTech Reserve Driftwood field decking, hidden fasteners
4. Picture frame + breaker boards — TimberTech Reserve Driftwood
5. Fascia

House shown for context only (the deck work only). The frame under the decking is called out on S-102 and A-003.

Axonometric · not to scale · the model the plans are drawn from

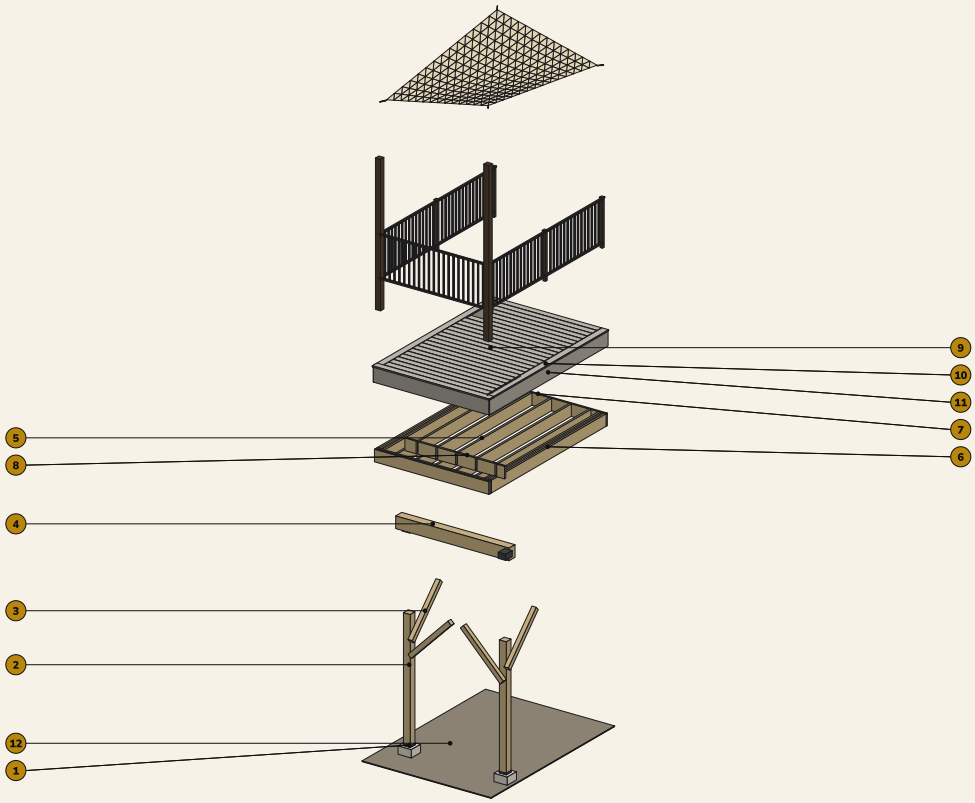
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-003 ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts + knee braces · beams · frame · decking · rail

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. TimberTech Reserve Driftwood field decking, hidden fasteners
10. Picture frame + breaker boards — TimberTech Reserve Driftwood
11. Fascia
12. Gravel over 6-mil poly under the decks

Layers lifted in build order. Lifts are for clarity only; every member returns straight down to its place.

Axonometric · not to scale · the model the plans are drawn from

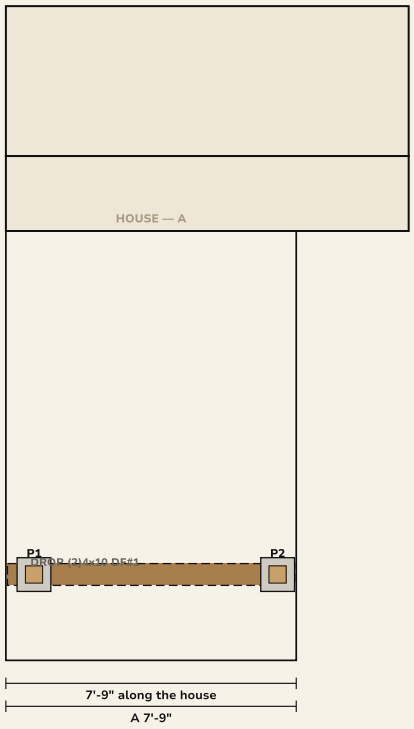
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	9"	9'-2"	A	2,214 lb
P2	7'-3"	9'-2"	A	2,214 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 2,214 lb

Plan · not to scale · house at the top · dimensions govern

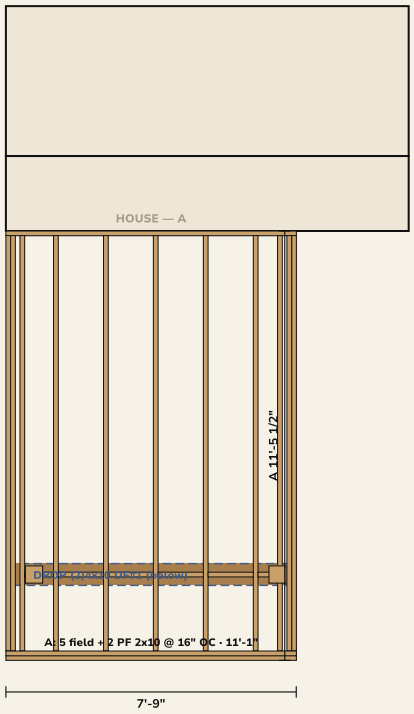
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRAMING NOTES

7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"

2-ply 2x10 front and sides

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

blocking one row over the beam, staggered

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Connectors:

1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post): 4

1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam): 2

1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block): 2

Simpson LUS28Z face-mount hanger 2x8/2x10: 14

Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 4

Simpson DTT1Z lateral tie kit w/ SDS screws: 4

Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden): 9

Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal:

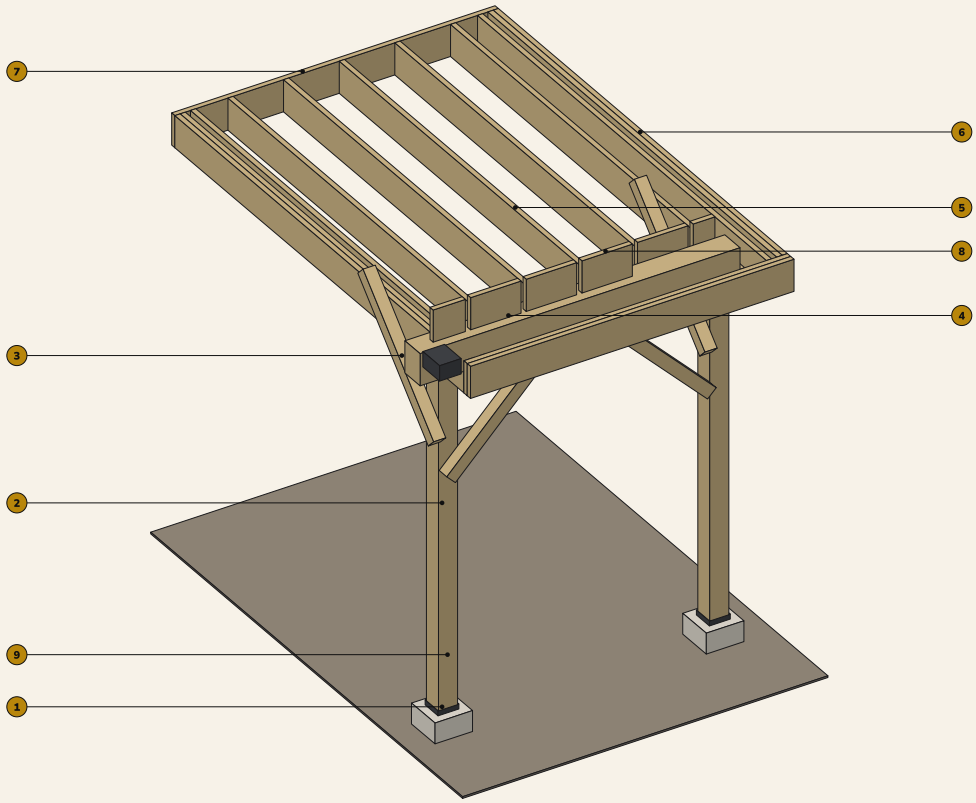
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-102 STRUCTURE ISOMETRIC — FRAME, LABELED

Piers, post bases, posts, knee braces, post caps, beam, joists, double rims, ledger, blocking

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. Gravel over 6-mil poly under the decks

Decking, rail and house removed to show the frame. Joists hang from the ledger in face-mount hangers, bear on the drop beam with H2.5AZ ties; double rims in concealed-flange hangers.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

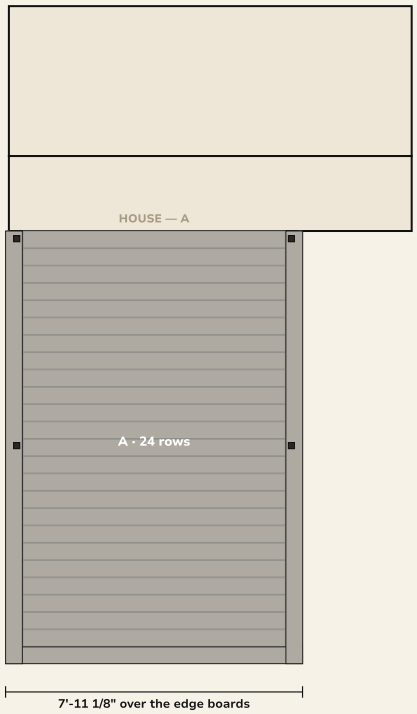
A-101 Decking plan

A-101 **DECKING PLAN**

Every board, border and divider as it lays; rail posts shown for the pockets

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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DECKING

Field: TimberTech Reserve Driftwood, 5.36" x 0.94", 3/16" gaps

Border: Reserve Driftwood

Boards run parallel to the house; edge overhang 1 1/16" over the fascia

CONCEALoc at every joist in every gap (168)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per zone: A 24 (no rips)

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

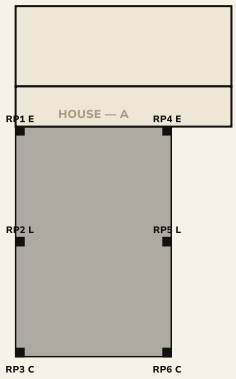
A-201 Railing plan + elevation

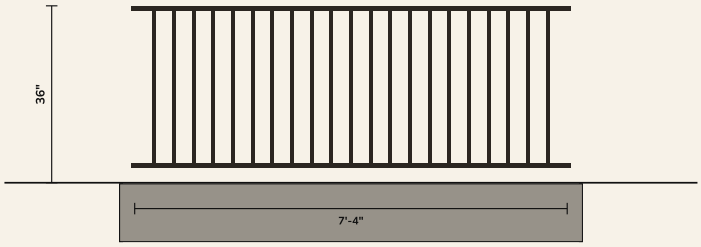
A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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RAILING

IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

Bays: left 5'-6 1/4" · left 5'-6 1/4" · right 5'-6 1/4" · right 5'-6 1/4" · front 7'-4"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.

Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

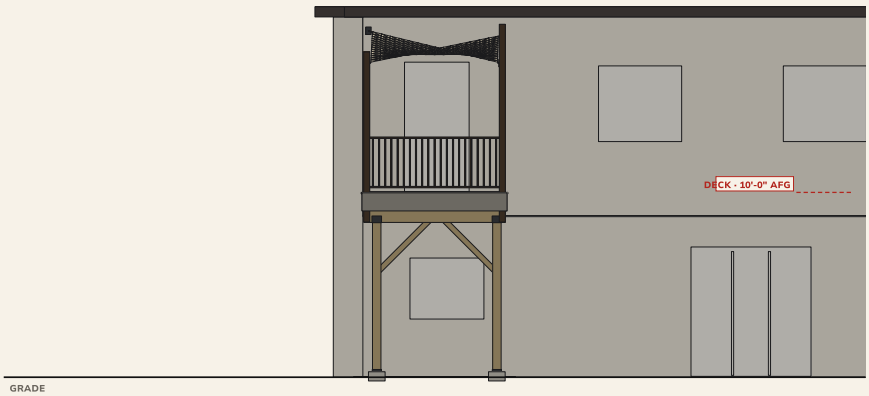
A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

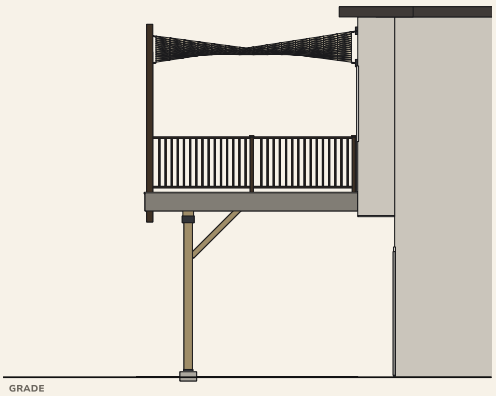
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRONT ELEVATION — from the yard



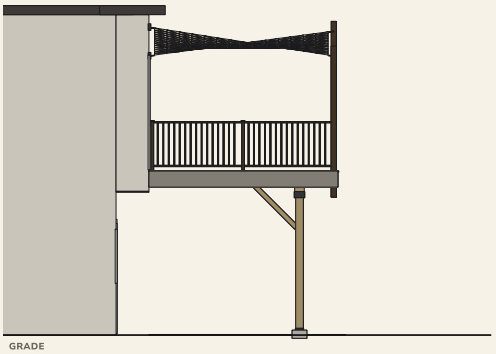
RIGHT END ELEVATION



NOTES

· Deck surface 10'-0" above grade
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Deck frame: 2x10 #1 True Frame GC @ 16" OC on a (2)4x10 DF #1 drop beam, 2 posts with 4x4 knee braces both ways
· Guard 36" min. on every open side — the surface is 10'-0" above grade (IRC R312.1)

LEFT END ELEVATION



Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-301 Section + front elevation

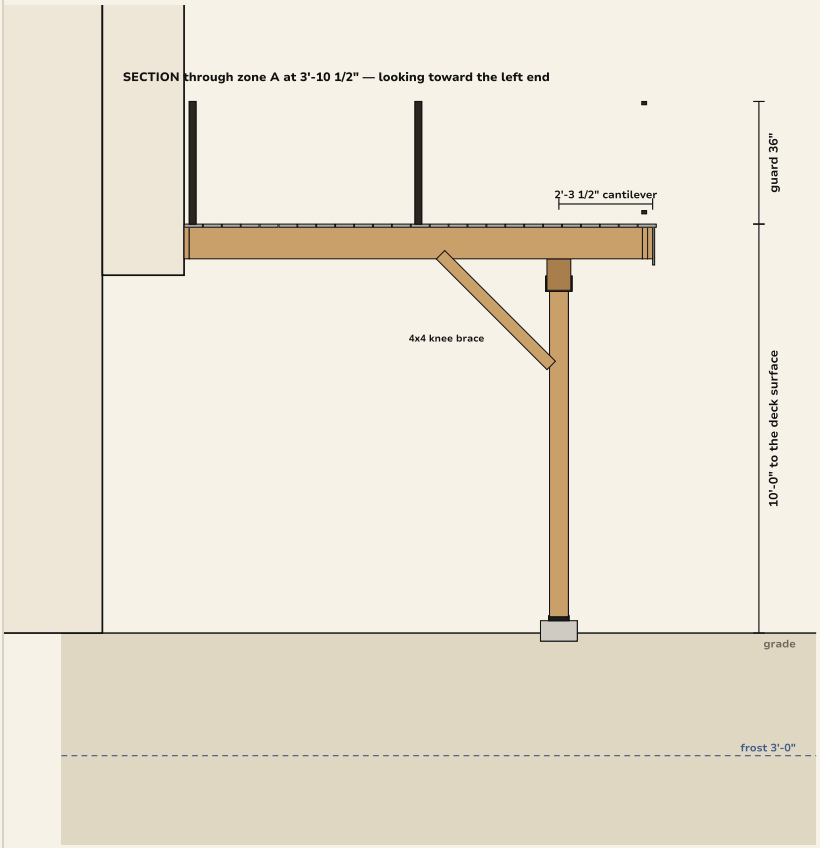
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

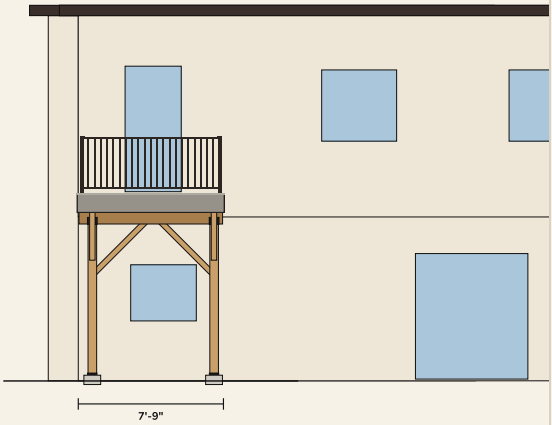
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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SECTION through zone A at 3'-10 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Posts 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

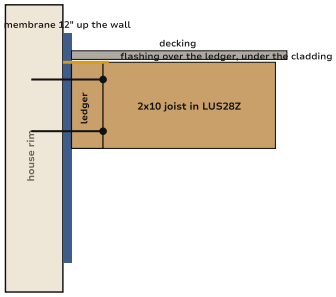
D-501 DETAILS

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

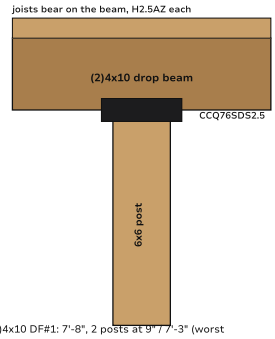
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1 LEDGER AT THE HOUSE



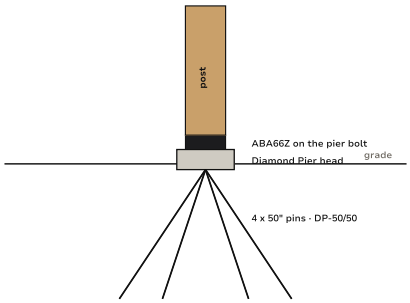
2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

2 POST · BEAM · CAP

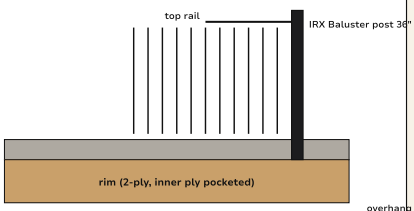


DROP (2)4x10 DF#1: 7'-8", 2 posts at 9' 7 7'-3" (worst 2,214 lb)

3 FOOTING · POST BASE

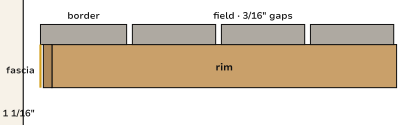


4 RAIL POST · GUARD



IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

5 DECK EDGE · BORDER



CONCEALoc at every joist in every gap (168)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.

Schematic details, not to scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

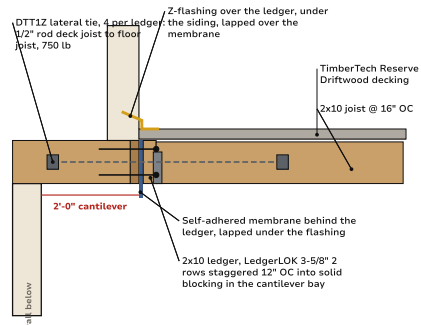
D-502 CONNECTION DETAILS — FRAME

Ledger on the cantilever · hangers · joists on the beam · (2)4x10 on the cap · pier + base · knee braces

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

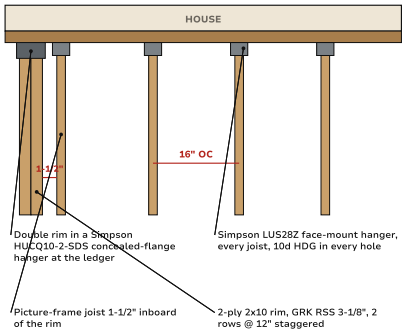
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1 LEDGER ON THE CANTILEVERED FLOOR — SECTION

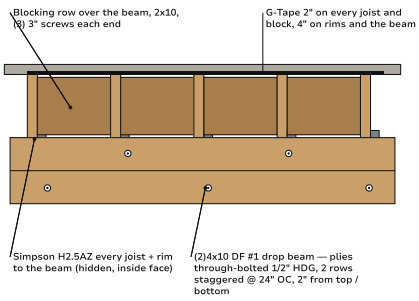


ENGINEER: Arapahoe County bars a ledger on a cantilevered floor without engineer approval. The stamped detail governs: ledger into the blocked cantilever joists, or a freestanding beam on posts at the house.

2 HANGERS — PLAN AT THE LEDGER + SIDE RIM



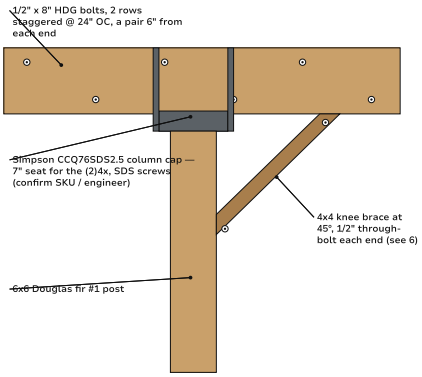
3 JOISTS ON THE DROP BEAM — SECTION



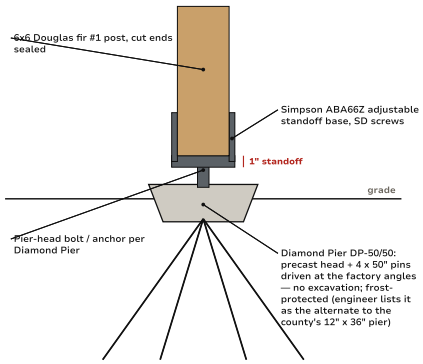
Joists bear the full beam width

Cantilever 2'-3 1/2" past the beam — over L/4: engineer

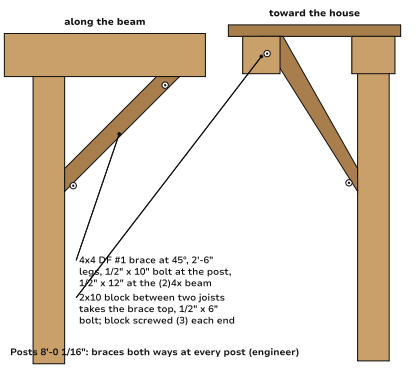
4 (2)4x10 BEAM ON THE 6x6 — ELEVATION



5 POST BASE ON THE DIAMOND PIER — SECTION



6 KNEE BRACES — BOTH DIRECTIONS



Posts 8'-0 1/16": braces both ways at every post (engineer)

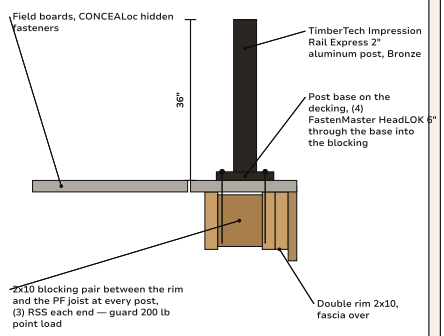
D-504 CONNECTION DETAILS — DECK EDGE, RAIL, LATERAL

Rail post mounting · picture frame + fascia · guard bay · DTT1Z lateral ties · joist tape · fastening schedule

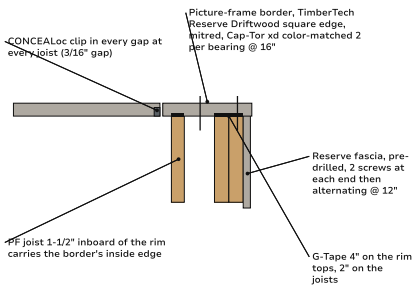
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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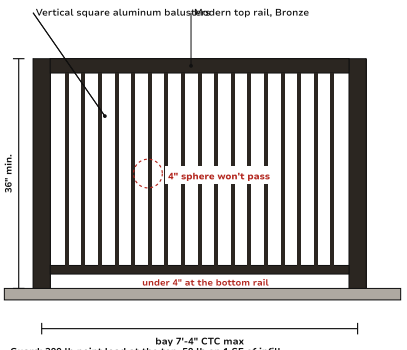
1 RAIL POST — SURFACE MOUNT ON BLOCKING



2 DECK EDGE — PICTURE FRAME + FASCIA

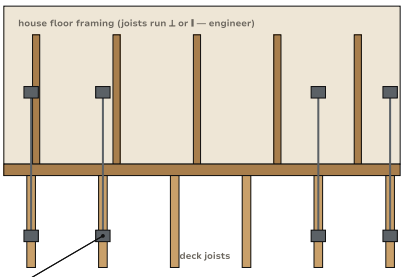


3 GUARD — ONE BAY, ELEVATION



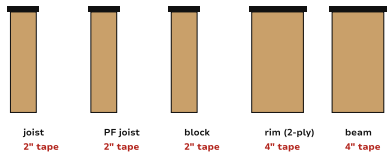
Guard: 200 lb point load at the top, 50 lb on 1 SF of infill (IRC Table R301.5) — submit the rail code report

4 LATERAL LOAD — DTT1Z TIES (PLAN)



(4) Simpson DTT1Z, 2 near each end of the ledger: 1/2" threaded rod through the ledger and the band into a floor joist (or blocking) inside — 750 lb each (IRC R507.9.2)

5 JOIST TAPE + BLOCKING

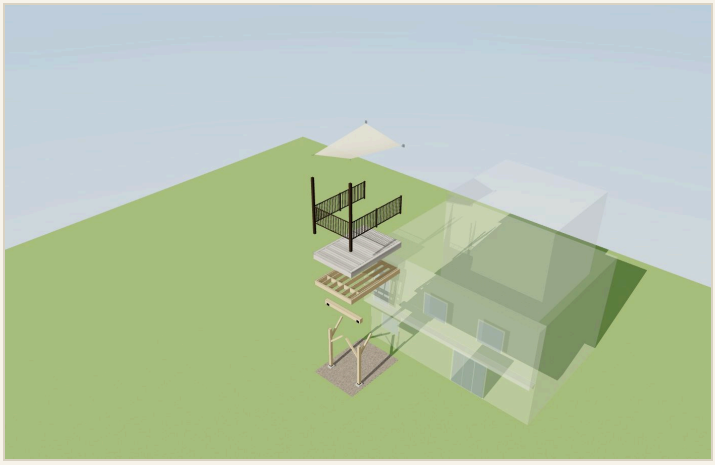


Nichigo G-Tape 3035 (black) on every top edge before decking, lapped over the ends and down the sides 1/2"; one row of 2x10 blocking over the beam, (3) structural screws each

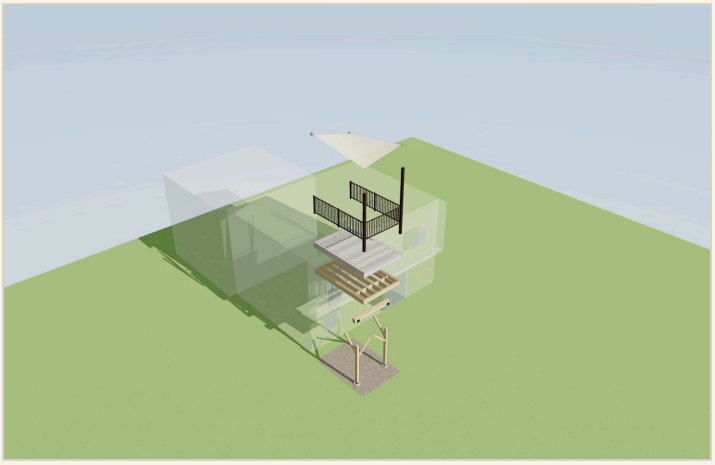
6 FASTENING SCHEDULE

Ledger to house	LedgerLOK 3-5/8", 2 rows staggered 12" OC — engineer
Lateral	(4) DTT1Z, 1/2" rod into the floor framing
Joist to ledger	LUS28Z, 10d HDG every hole
Rim to ledger	HUCQ10-2-SDS, SDS screws
Rim plies	GRK RSS 3-1/8", 2 rows @ 12" staggered
Joist to beam	H2.5AZ each joist + rim, 10d x 1-1/2"
Beam plies	1/2" x 8" HDG bolts, 2 rows staggered @ 24"
Beam to post	CCQ76SDS2.5 cap, SDS screws
Post to pier	ABA66Z on the Diamond Pier, SD screws
Knee braces	1/2" through-bolt each end (x10 / x12 / x6)
Field boards	CONCEALoc, every joist, every gap
Borders / fascia	Cap-Tor xd color-matched, pre-drilled
Rail posts	(4) HeadLOK 6" into blocking

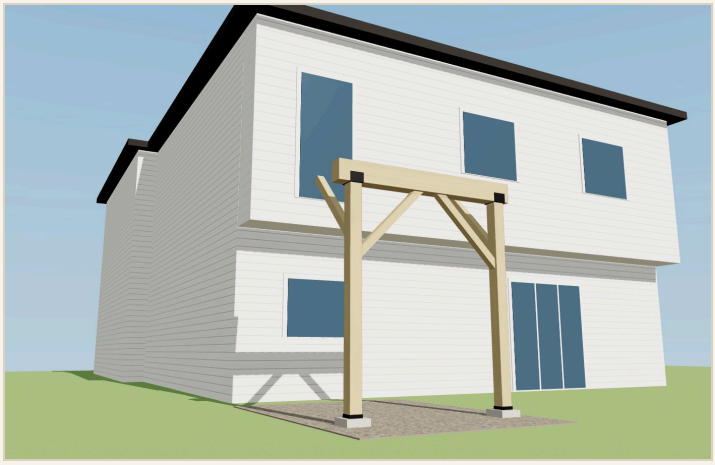
EXPLODED VIEWS



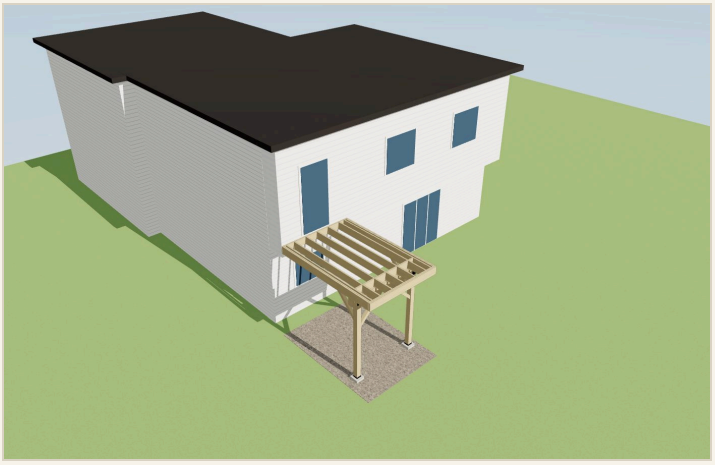
Exploded — every layer



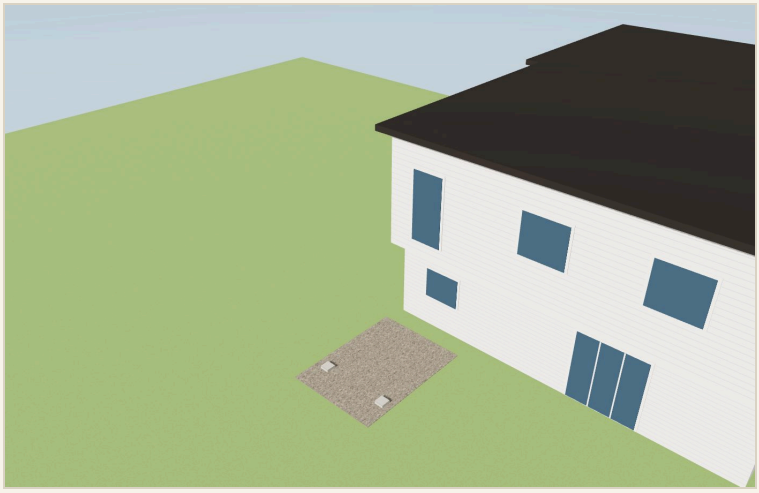
Exploded — from the open corner



Posts and beams on the footings



Frame only — joists, rims, beams, ledger



Step 1 · Footings — 2 Diamond Pier DP-50/50 driven-pin footings

T-400 · Step 1 · Footings

GOES IN

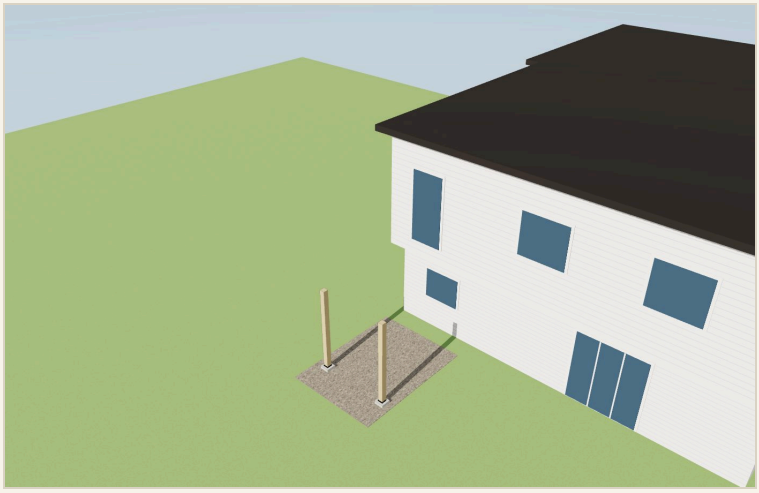
- (2) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: DROP (2)4x10 DF#1 at 9'-2" from the reference wall, posts at 9" / 7'-3".
- Set 2 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 Douglas fir #1 on ZMAX bases

T-401 · Step 2 · Posts

GOES IN

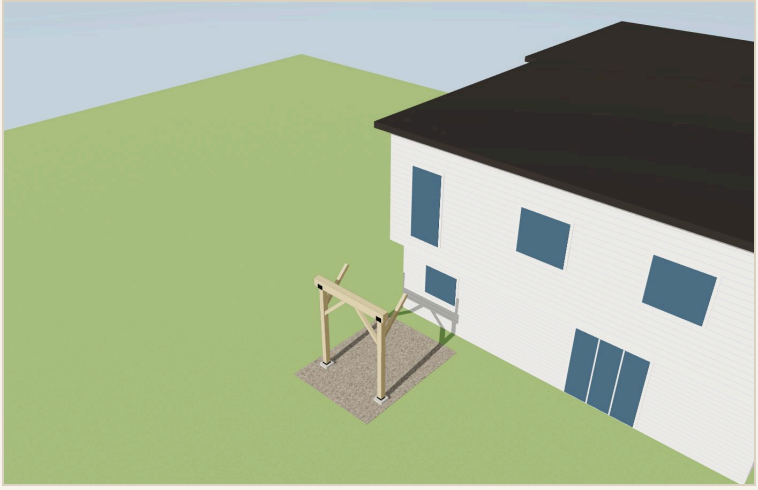
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt
- (1) Sail hardware: 4 stainless turnbuckles (jaw-jaw), D-shackles, snap hooks, 2 house anchor plates with through-bolts / structural screws into the framing

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): DROP (2)4x10 DF#1 posts 8'-0 1/16" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff



Step 4 · Beams — DROP (2)4x10 DF#1 (2 posts)

T-403 · Step 4 · Beams

GOES IN

- (2) 4x10x8 Douglas Fir #1
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal
- (2) 4x4x8 Douglas Fir #1 — knee braces
- (4) 1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)
- (2) 1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)
- (2) 1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)
- (10) 1/2" x 8" HDG carriage bolt + nut + washer

HOW

- DROP (2)4x10 DF#1: 7'-8" one piece, 2 posts @ 6'-6" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.
- Two-ply beam: through-bolt the plies with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a pair 6" from each end — before it goes up.
- Knee braces once the beam is capped: 4x4 at 45°, 2'-6" legs, one along the beam toward mid-span and one toward the house up to a 2x10 block between two joists (set the block with the frame) — 1/2" through-bolt each end, then pull the temporary bracing.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened
- Every knee brace bolted both ends; posts still plumb



Step 5 · Frame — 2x10 joists at 16", rims, ledger, blocking over the beam

T-404 · Step 5 · Frame

GOES IN

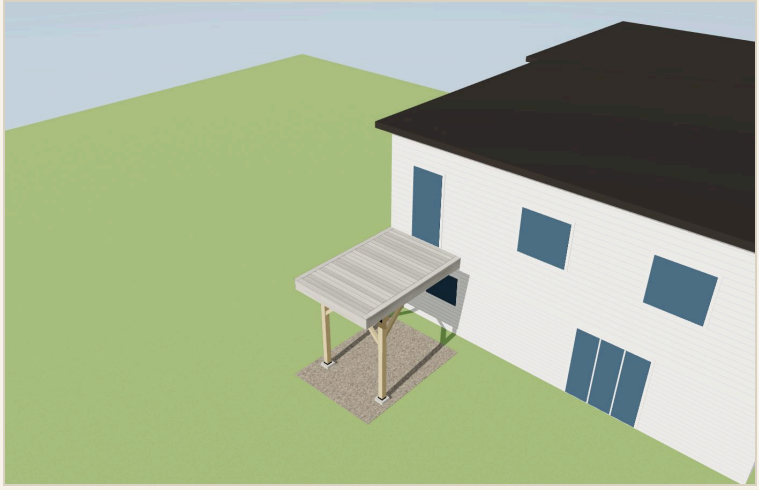
- (15) 2x10x12 #1 True Frame Joist SYP GC
- (14) Simpson LUS28Z face-mount hanger 2x8/2x10
- (4) Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)
- (1) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (1) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (9) Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)
- (1) Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams
- (2) Nichigo G-Tape 3035 2" x 65' black

HOW

- 2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
- A: 5 joists @ 16" OC x 11'-1". Crown up, hangers at the ledger.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 6 · Decking — TimberTech Reserve Driftwood, hidden fasteners

T-405 · Step 6 · Decking

GOES IN

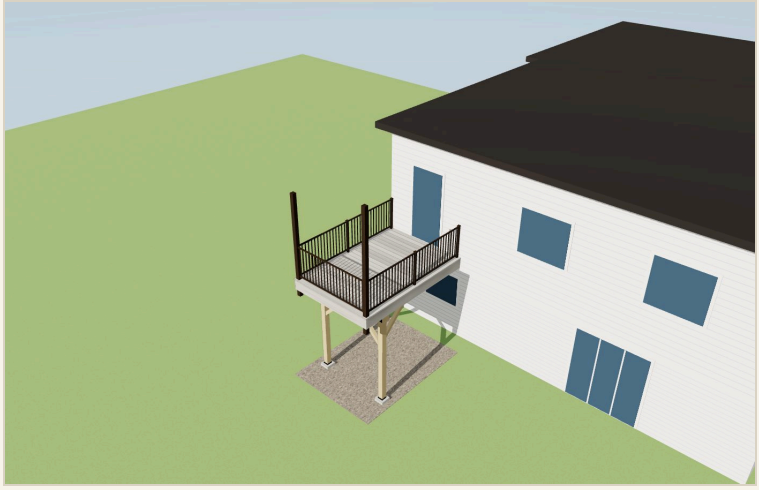
- (12) TimberTech Reserve Driftwood 1x6x16 Grooved
- (1) TimberTech Reserve Driftwood 1x6x12 Square Edge — borders
- (1) TimberTech Reserve Driftwood 1x6x20 Square Edge — borders
- (3) TimberTech Reserve Fascia Driftwood 0.58x11.95x12
- (1) TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)
- (1) Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct

HOW

- CONCEALoc at every joist in every gap (168)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
- Rows per zone: A 24. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · Rail — IRX Baluster 36" Bronze

T-406 · Step 7 · Rail

GOES IN

- (4) TimberTech Impression Rail Express 6' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze
- (1) TimberTech Impression Rail Express 8' x 36" level kit, vertical square aluminum balusters, modern top rail, Bronze
- (4) TimberTech Impression Rail Express 2" aluminum post kit 36" (post, base, cap, skirt), Bronze
- (1) TimberTech Bronze touch-up paint (cut ends)

HOW

- IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction

3D MODEL



The interactive 3D model lives at the online version of this document.

Quantities from the drawings; the stamped engineering set governs structure. Quote requested — no pricing on this sheet.